

Reaction types in Organometallic Chemistry

—Elementary Reactions in Organometallic Chemistry (基元反应)

- (1) Ligand coordination and substitution ✓
(配体的配位和取代)
- (2) Oxidative addition and reductive elimination ✓
(氧化加成和还原消除)
- (3) Insertion and elimination (de-insertion) reactions ✓
(插入反应和消除(反插入)反应)
- (4) Electrophilic and Nucleophilic Addition and Abstraction
(亲电和亲核加成与攫取)
(Reactions on the complexed ligands 和金属结合的配体的反应)

Help us to understand and predict the metal-catalyzed or mediated reactions.

Be aware:

- Classified **in terms of reaction patterns**, not mechanism !!
- An elementary reaction may go through different mechanisms.

Chapter 8. Electrophilic and Nucleophilic Addition and Abstraction (Reactions of the complexed ligands)

Outline

- A. Introduction**
- B. Nucleophilic addition to CO and CNR**
- C. Nucleophilic addition to olefins and acetylenes**
- D. Nucleophilic abstraction**
- E. Electrophilic addition**
- F. Electrophilic abstraction**

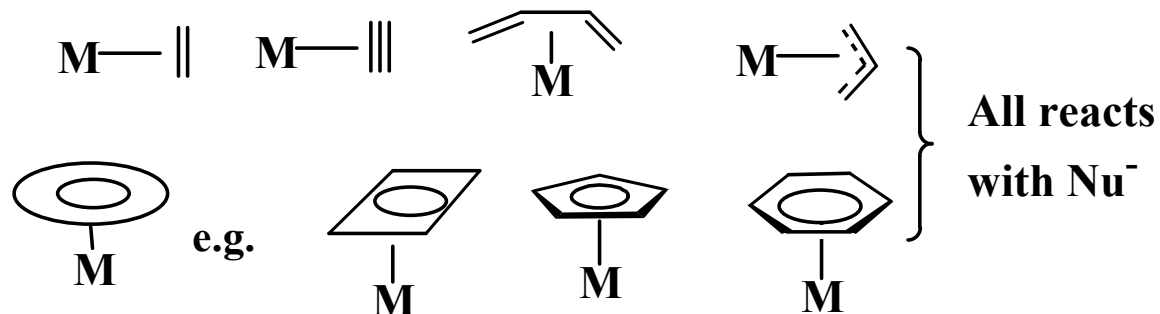
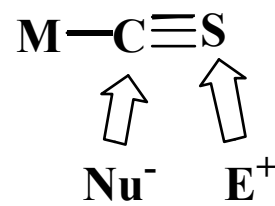
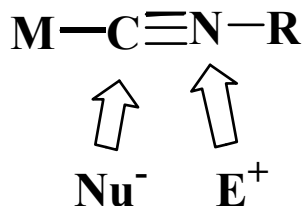
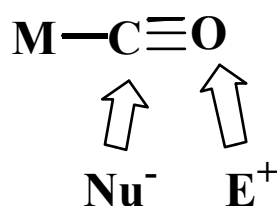
References and suggested readings:

1. *The Organometallic Chemistry of the Transition Metals*, Robert H. Crabtree, 6th Edition, 2014, Chapter 8.
2. *Organotransition Metal Chemistry*, Akio Yamamoto, 1986. 272-286

A. Introduction

Organic ligands are activated upon complexation. They may react with electrophiles or nucleophiles.

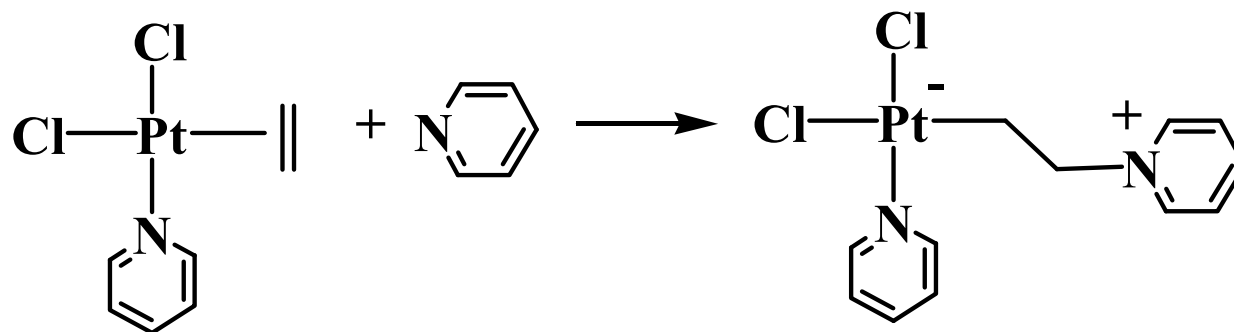
e.g. We have shown before:



Two possible outcomes for the reactions with nucleophiles

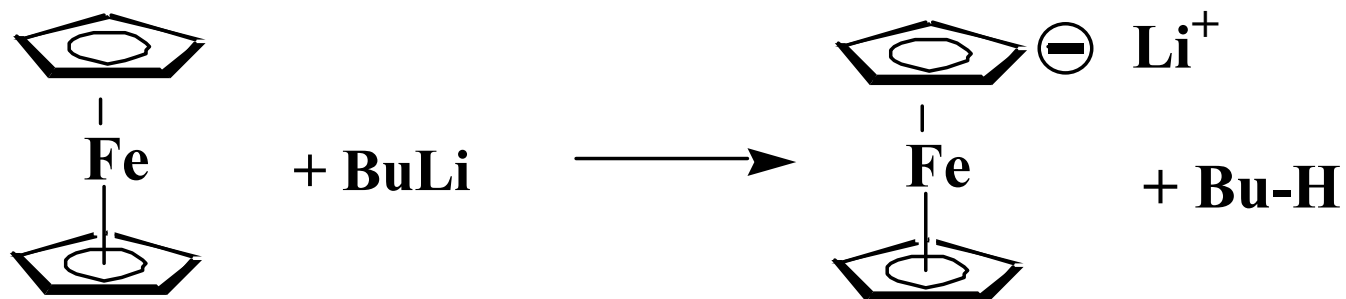
(1) *Nucleophilic addition* - reagent stays on metallic fragment

e.g.



(2) *Nucleophilic abstraction* (攫取) - reagents do not stay with metallic fragment

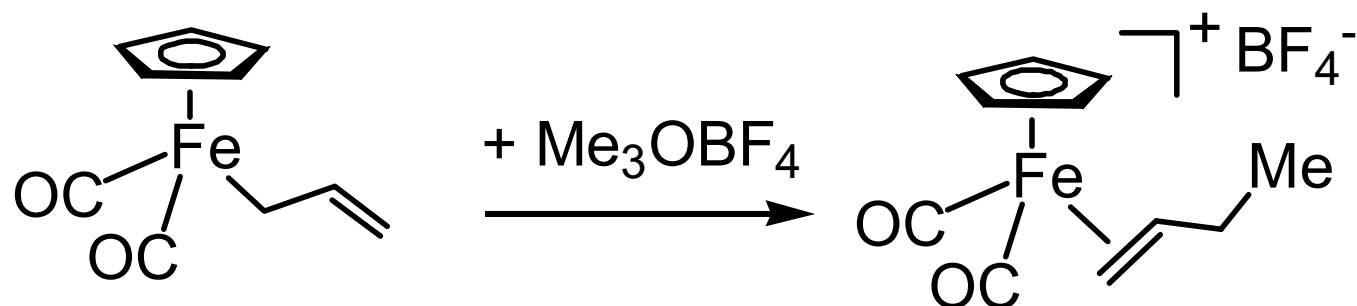
e.g.



There are also two possible outcomes for reactions with electrophiles.

(1) *Electrophilic addition* - reagent stays with metallic fragment.

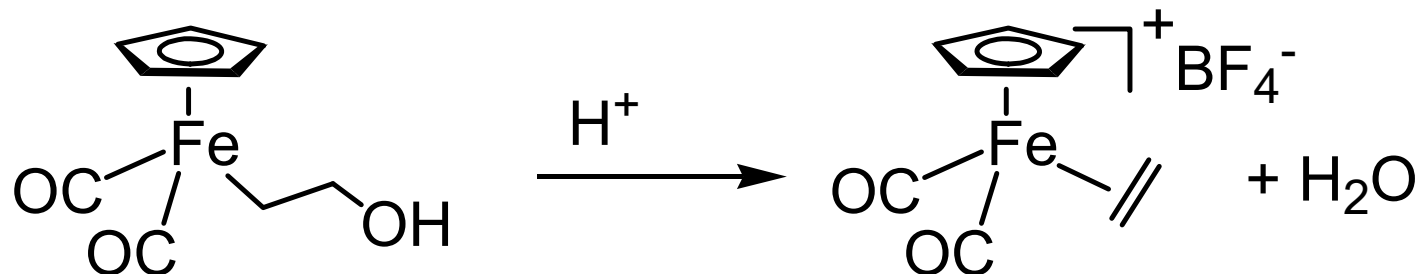
e.g.



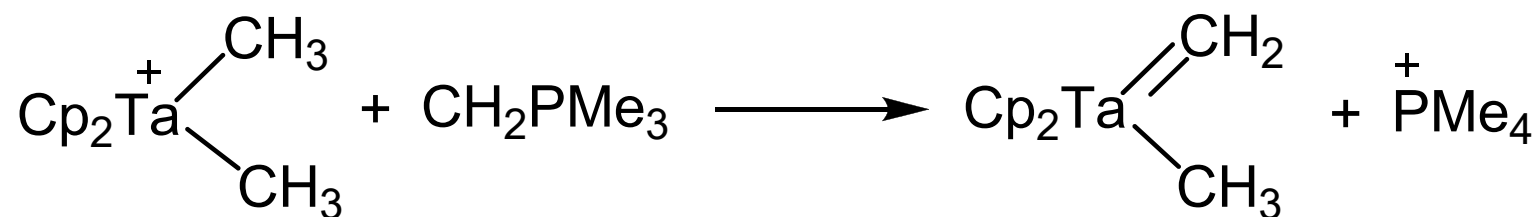
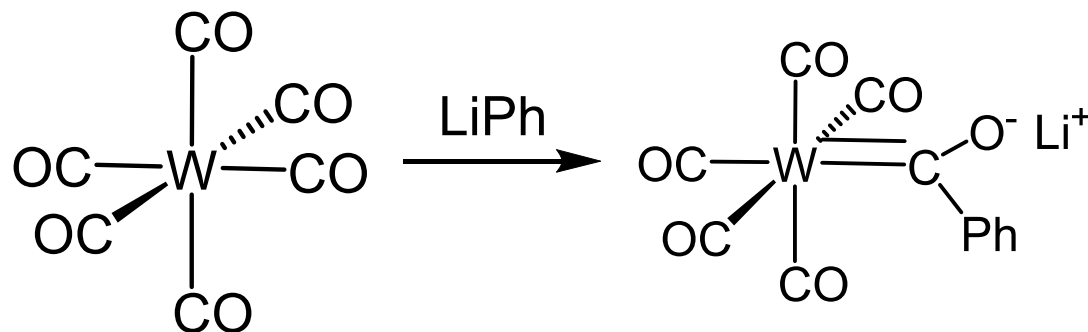
(Me_3OBF_4 is used to supply Me^+ .)

(2) *Electrophilic abstraction* - reagent does not stay with metallic fragment.

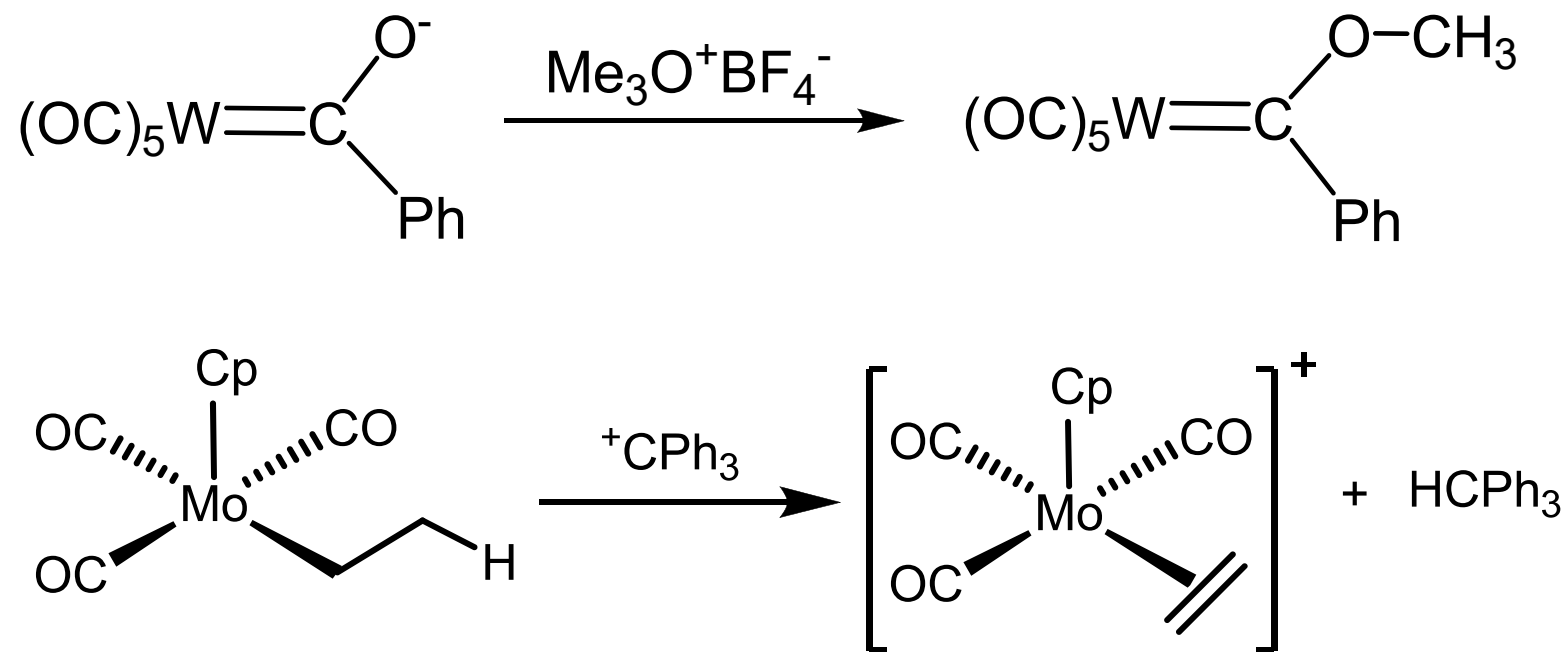
e.g.



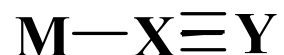
Nucleophilic attack is favored for complexes with net positive charges or with π -accepting (π -acid) ligands.



Electrophilic attack is favored for electron rich metal complexes, *i.e.* low valent metals, or those with net negative charges and/or those with strong electron donating ligands.

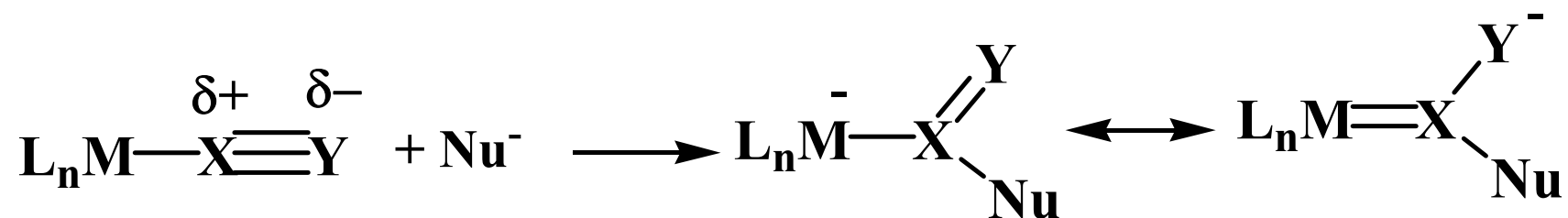


B. Nucleophilic addition to CO and related ligands



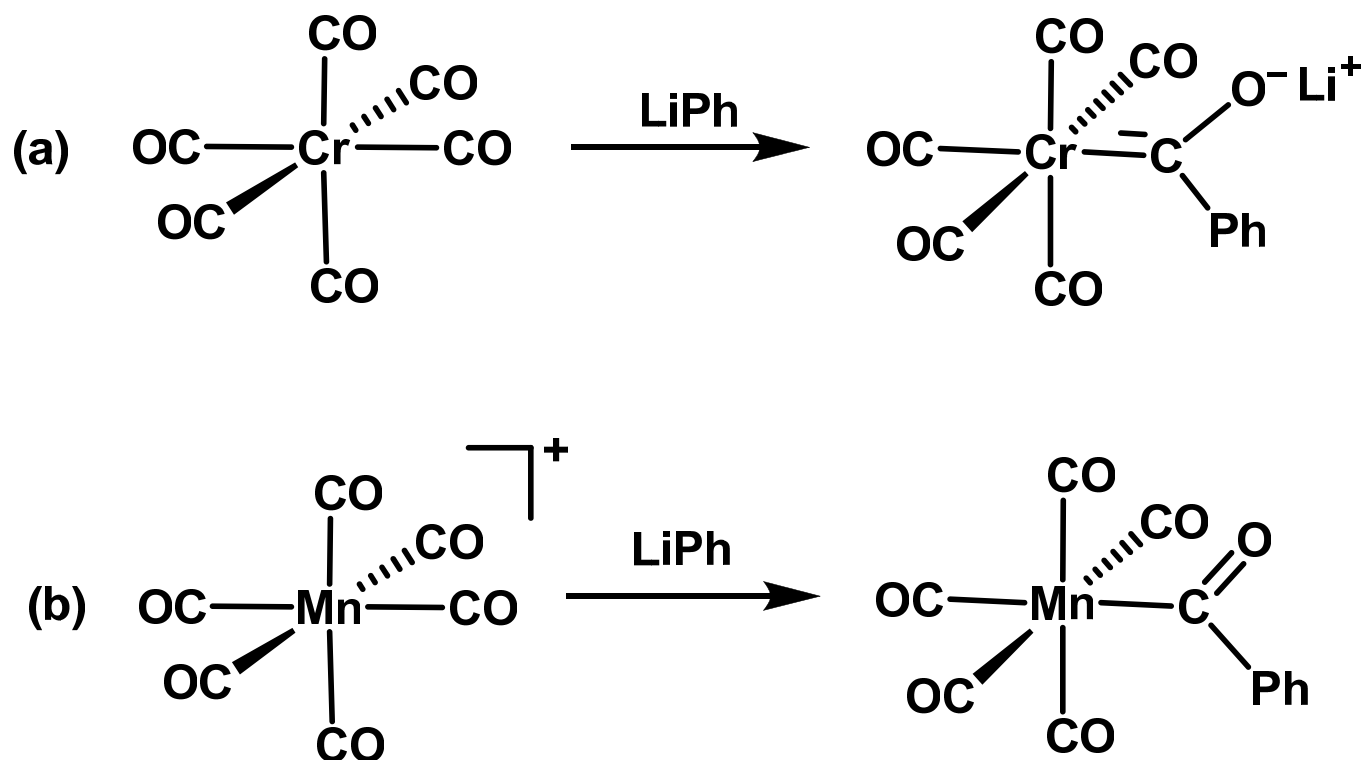
We have mentioned before that *M-CO*, *M-CNR* and *M-CS* can be attacked by Nu^- (at the α -position), especially positively charged species.

In general:



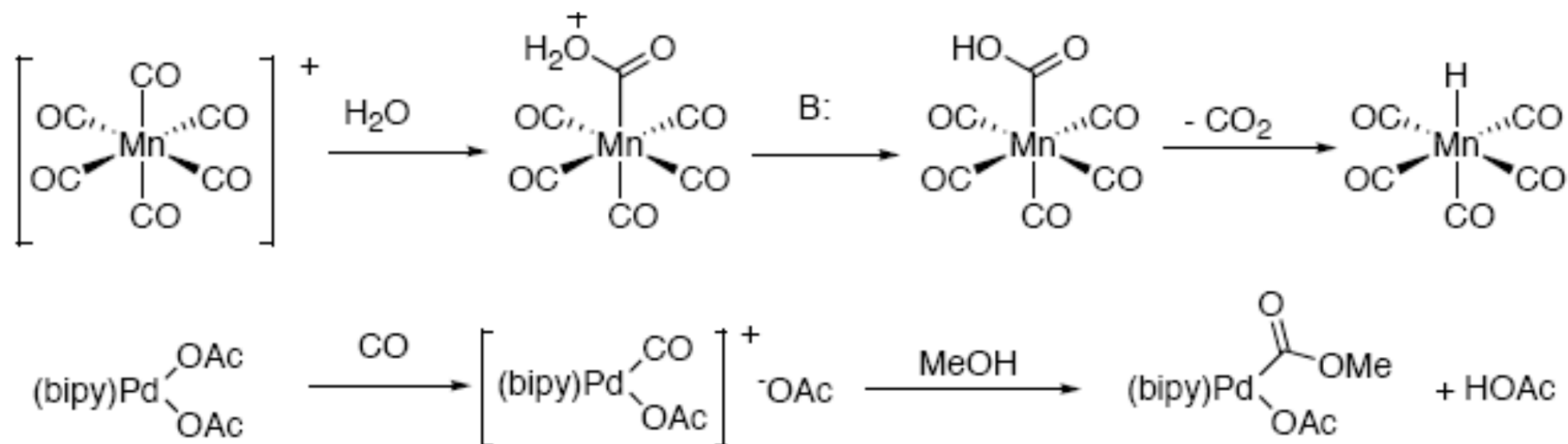
Review questions

Q1. Consider the reactions below, which one would you expect to be faster?

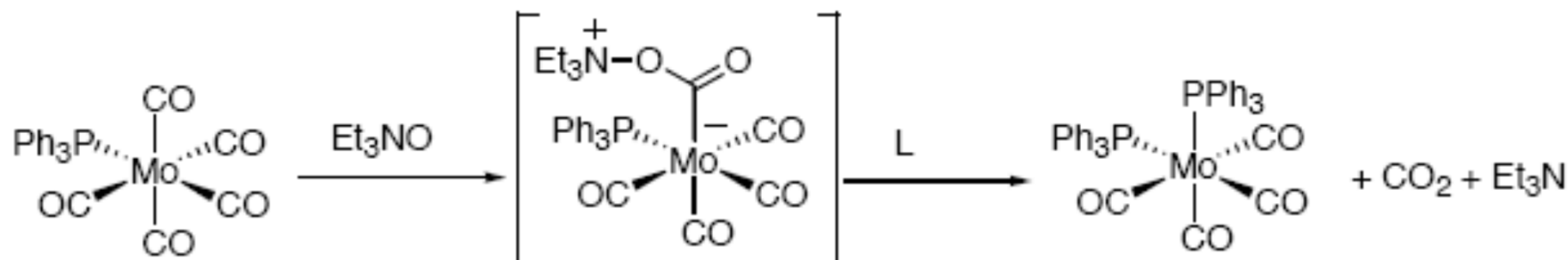


Answer: (b) (Positive charge species)

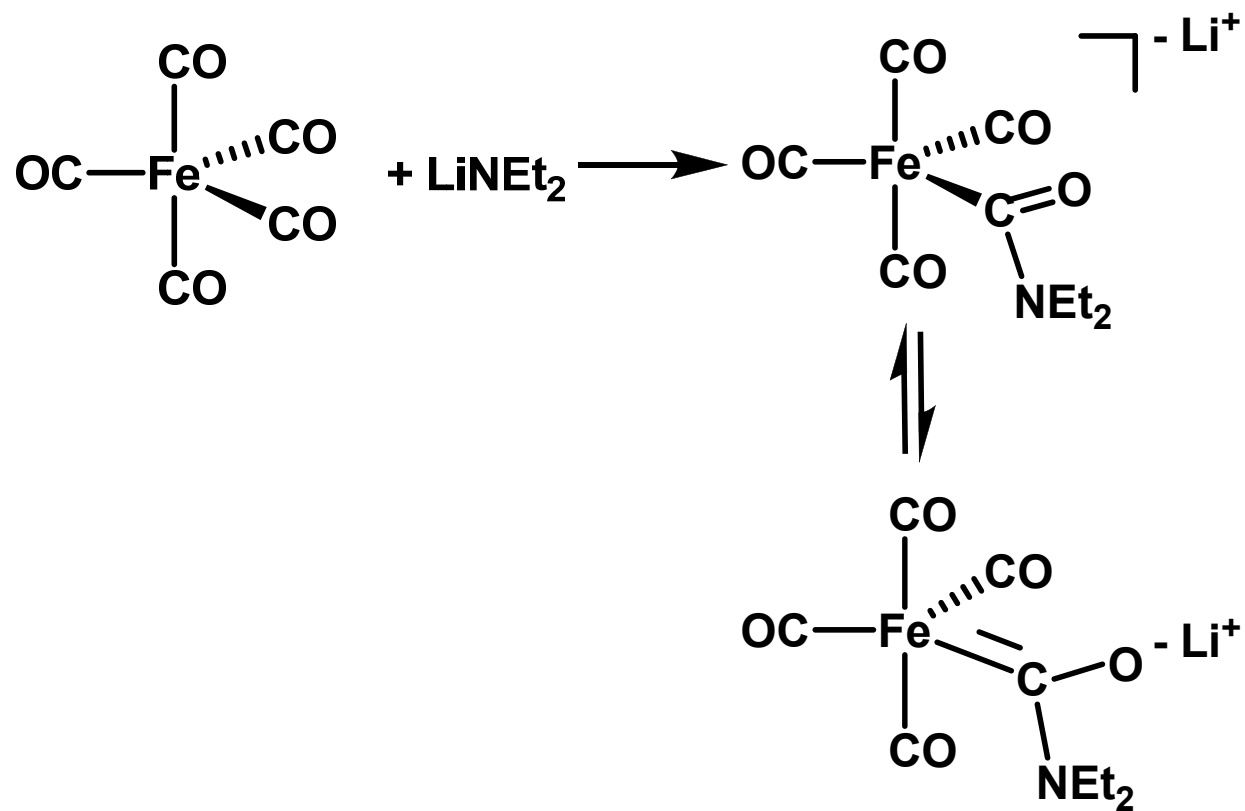
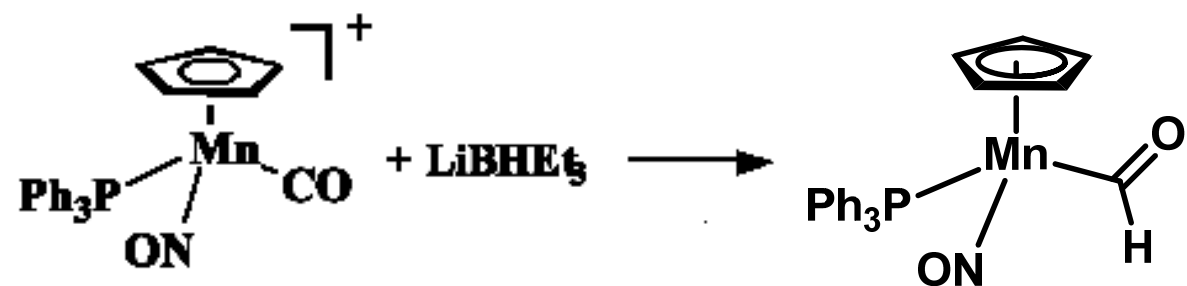
Increasing the electrophilicity of the metal center allows weaker nucleophiles like water or alcohols to attack.

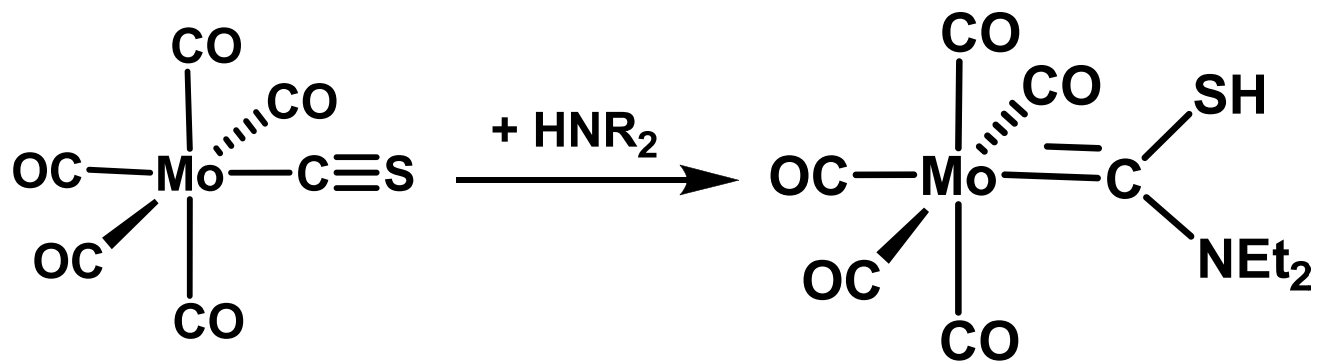
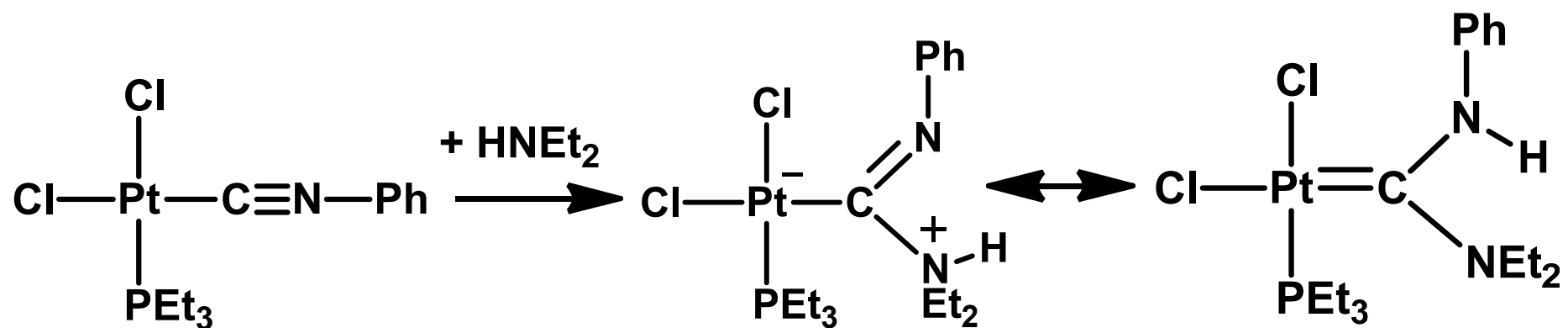


As discussed before, *N-oxides are strongly nucleophilic reagents that can be used to abstract CO ligands. The ligand substitution occurs exclusively cis to the phosphine.*

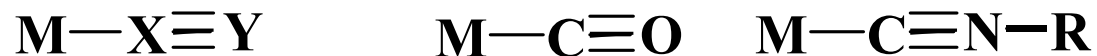


Q2. Provide the product of the following reactions





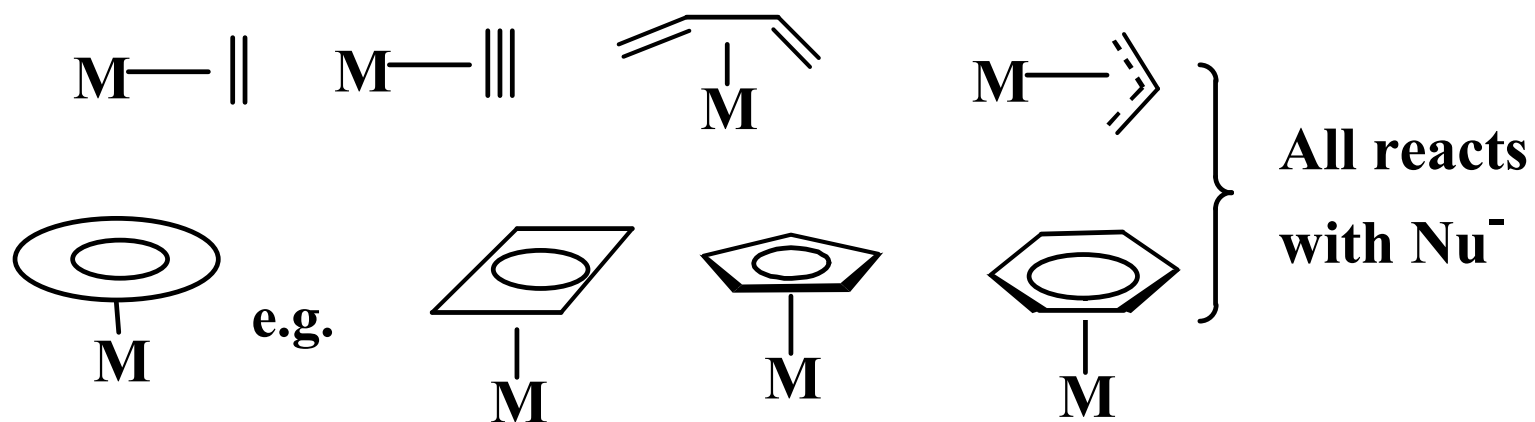
In terms of chemical properties for CO and related complexes:



We have mentioned before: *they are very similar.*

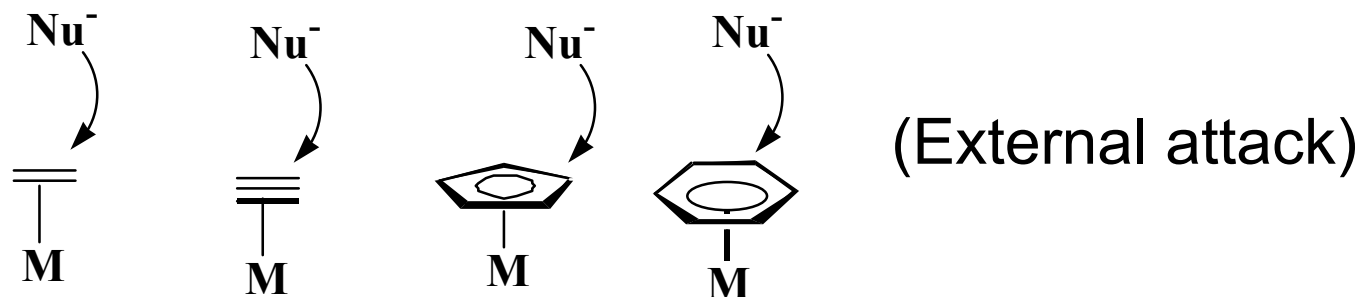
- ◆ M-CNR complexes: **more reactive** toward both nucleophiles and electrophiles than CO complexes.
- ◆ M-CS complexes: **very similar to M-CNR complexes**, more reactive than CO complexes.
- ◆ M-NO complexes: **less reactive** towards nucleophiles and electrophile than CO complexes.

C. Nucleophilic addition to coordinated olefins and acetylenes



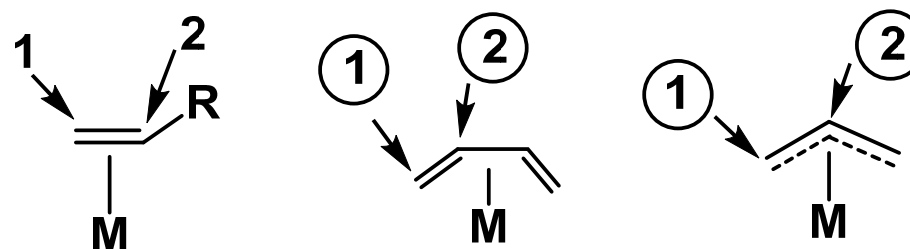
Most common features

- Direction of attack?** Attack from the opposite side of M



- Regio chemistry?**

e.g. attack site 1 or 2?

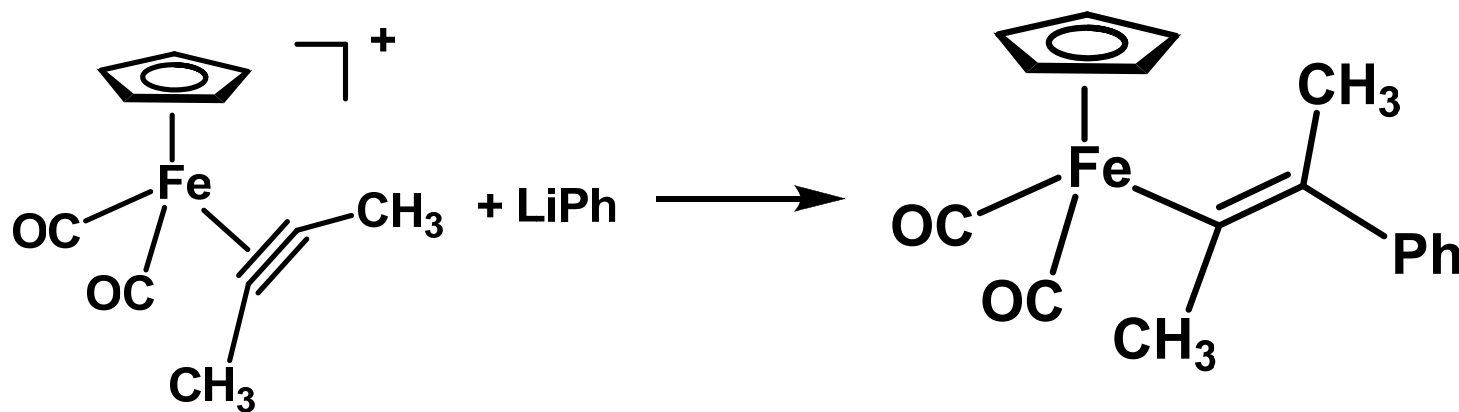


Attacked site

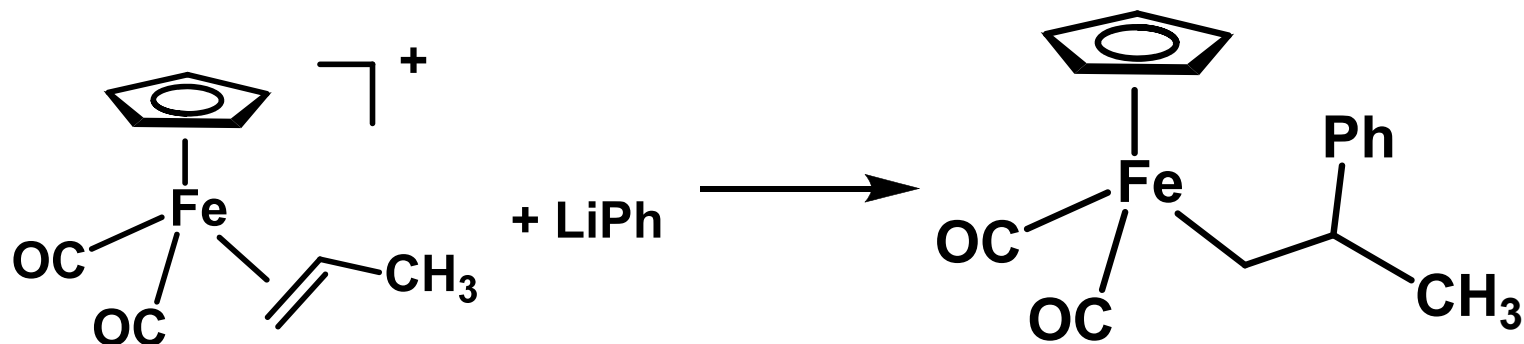
	Attacked site		
Preferred			
Less common			

Practice Problems

Give the products of the following reactions.

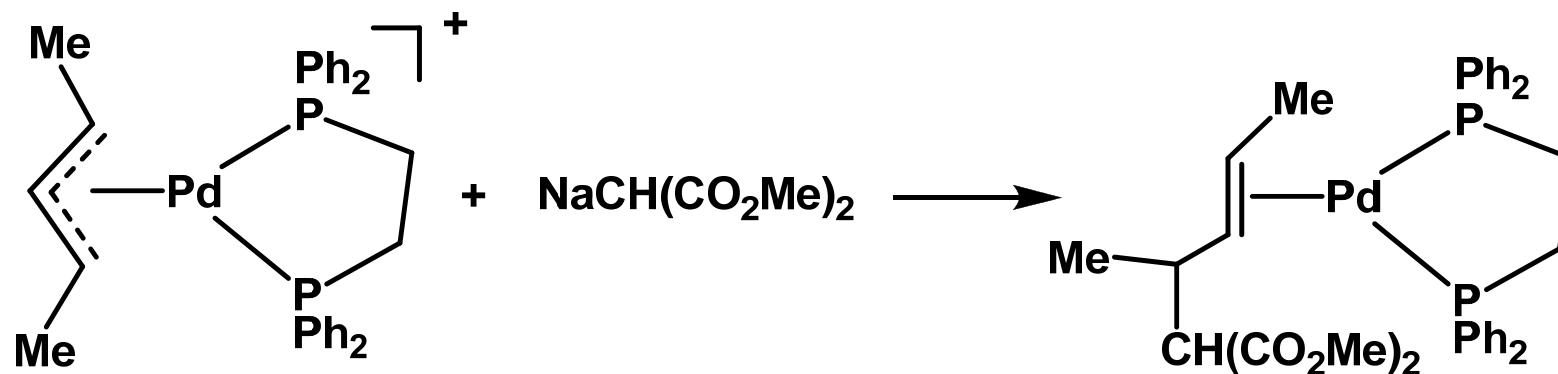
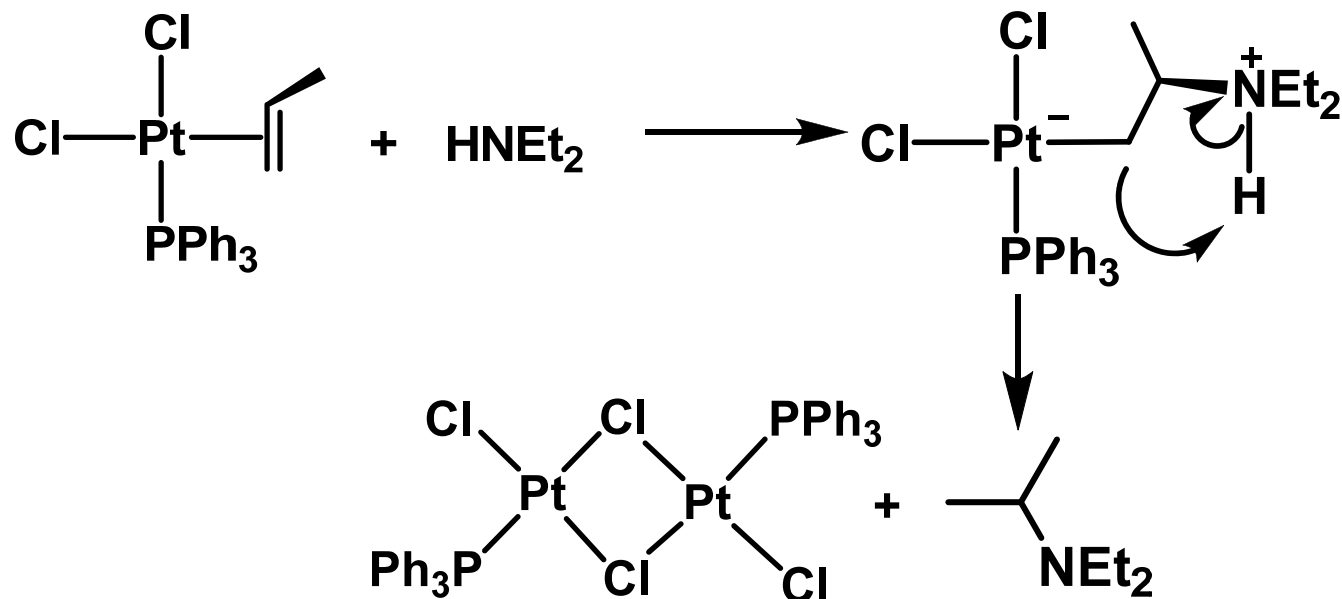


The stereochemistry is *syn* for the two substituents of $\text{C}\equiv\text{C}$.
(different from the case in *Insertion Reaction*)



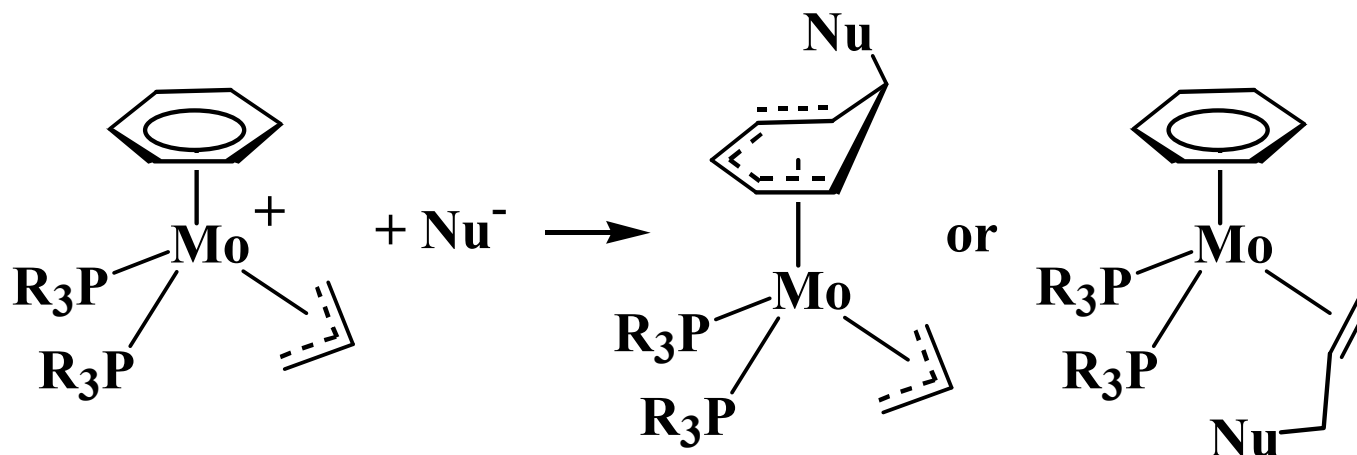
Practice Problems

Give the products of the following reactions.



• Site of selectivity.

If there are more than one type of olefins bonded to a metal, which one will be attacked? *e.g.*



There are several rules: Davies-Green-Mingos Rules

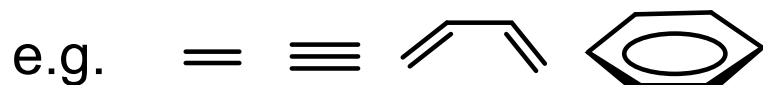
In a complex with several polyene or polyenyls, we often see selective attack at one site only. Davies, Green, and Mingos systematized these reaction outcomes in terms of rules that usually correctly predict the site of addition.

(经验规则，通常能正确地预测加成反应发生的位置)

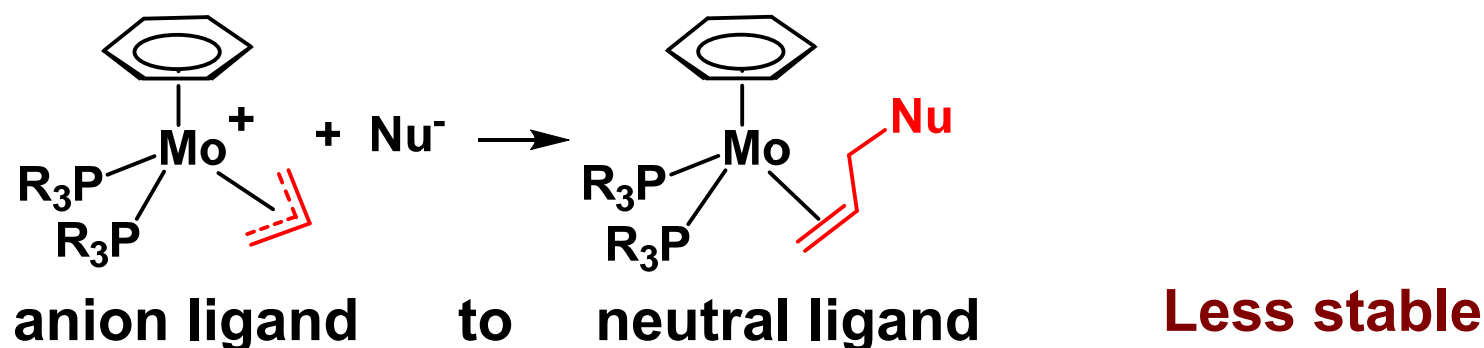
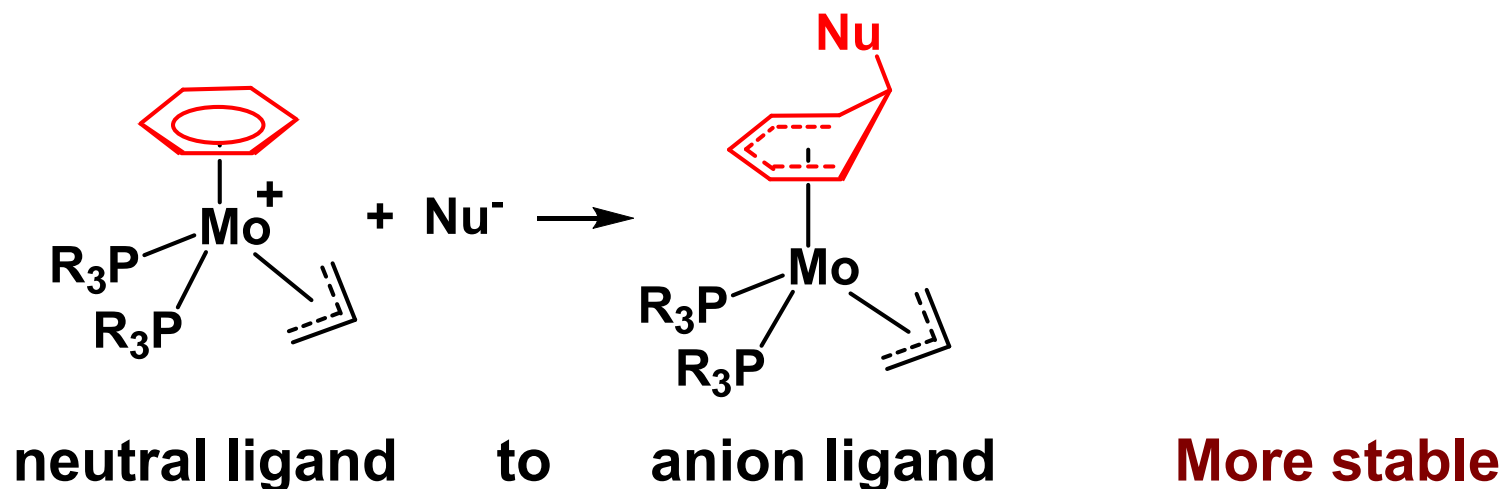
Davies-Green-Mingos Rules

Rule 1. Polyenes reacts before polyenyls

Polyenes (even # C)

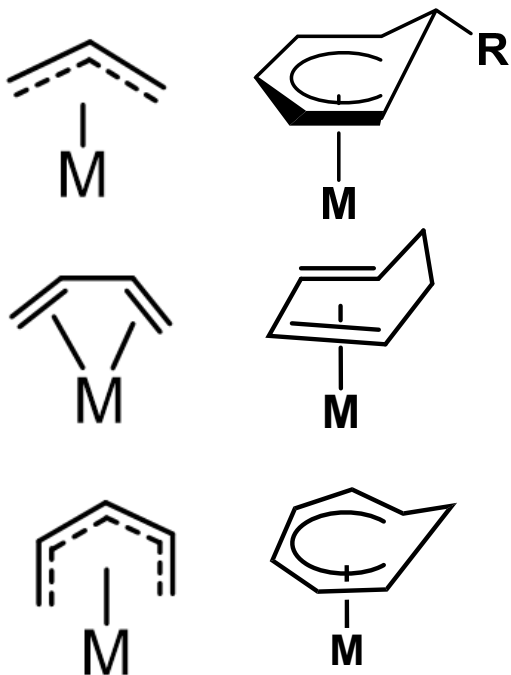


Polyenyls (odd # C)

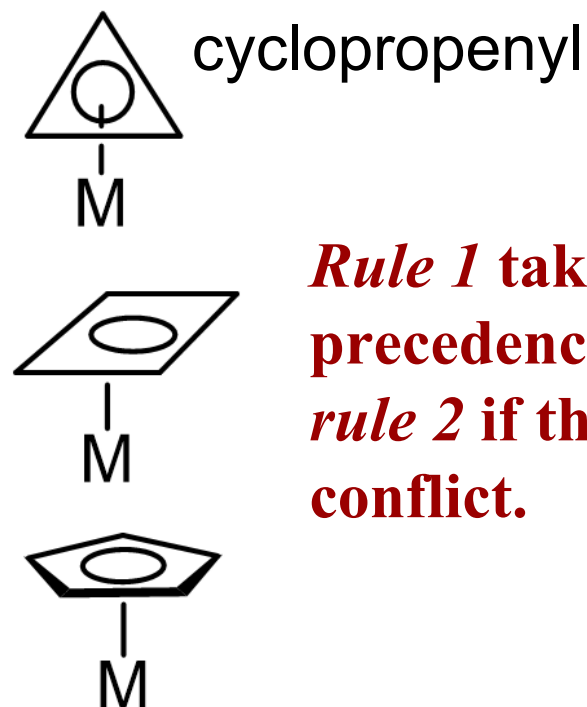


Rule 2. Open ligand reacts before closed ligands

Open ligand
(with interrupted
conjugation)

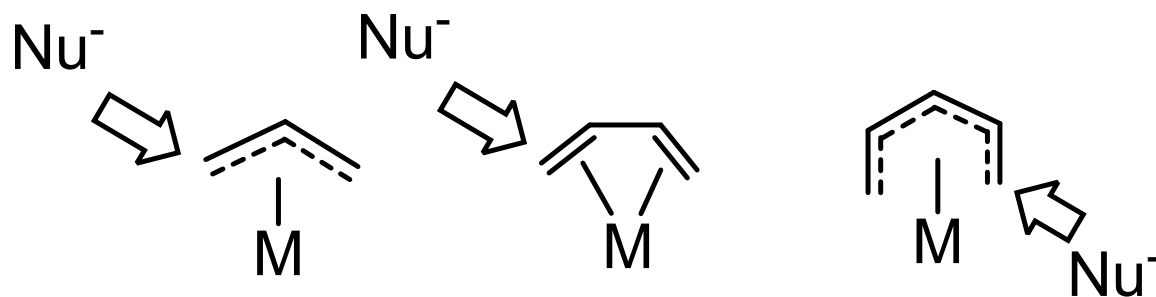


Closed ligand
(with cyclic conjugation)



***Rule 1 takes
precedence over
rule 2 if they
conflict.***

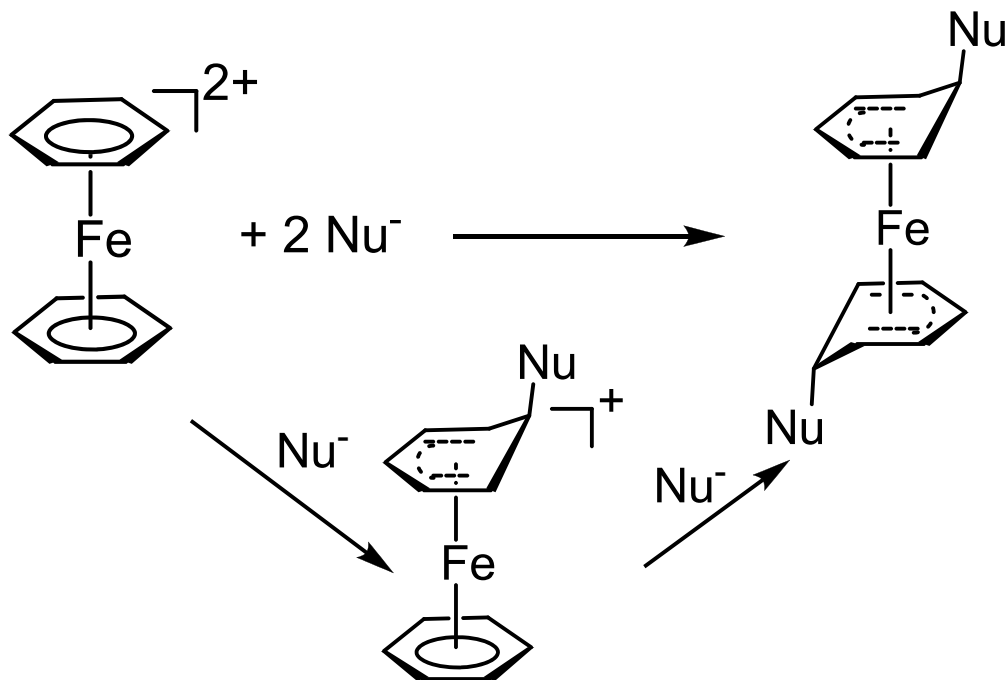
Rule 3. Usually terminal addition



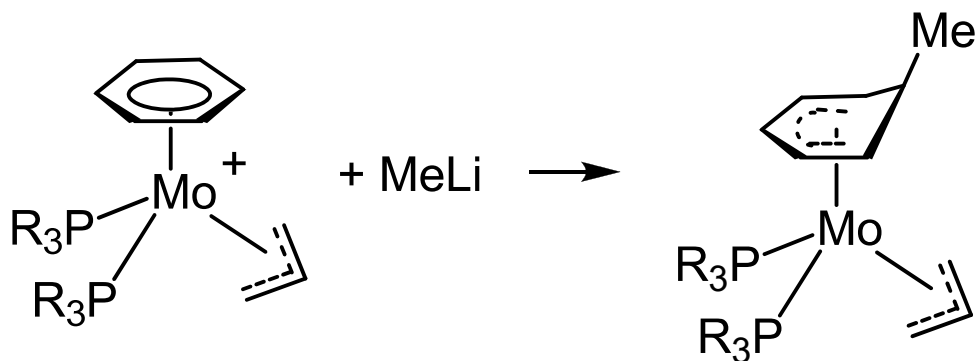
- Open polyenes give terminal addition.
- Open polyenyls usually give terminal attack, but non-terminal if L_nM is particularly electron donating.

Practice Problems

Give the products of the following reactions.

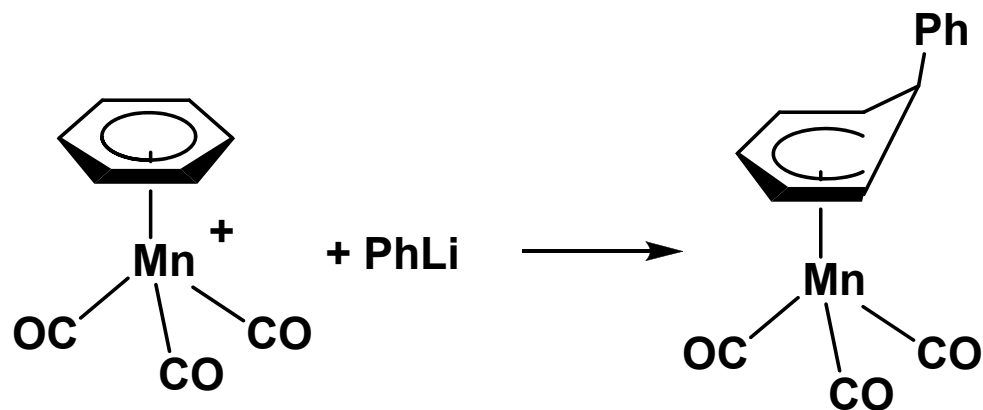
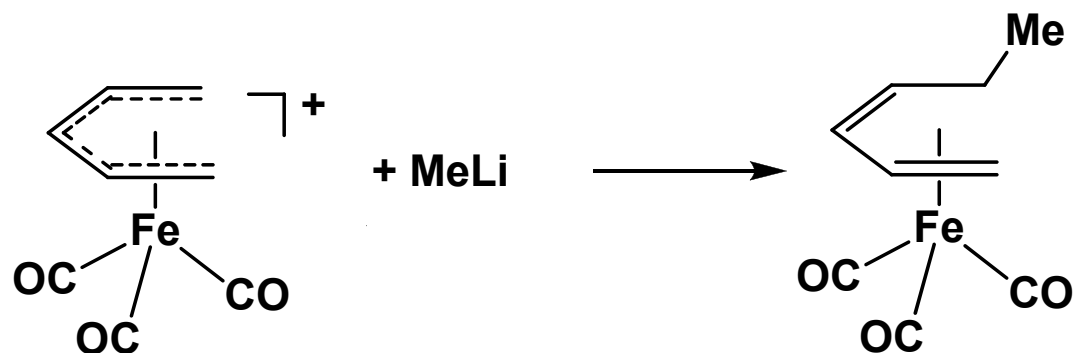


A second nucleophile can also add, but to the other ring, as predicted by Rule 1.



First attack the even closed arene rather than the odd open allyl

CO is an even ligand but, not being π -bound, is among the least reactive of these.

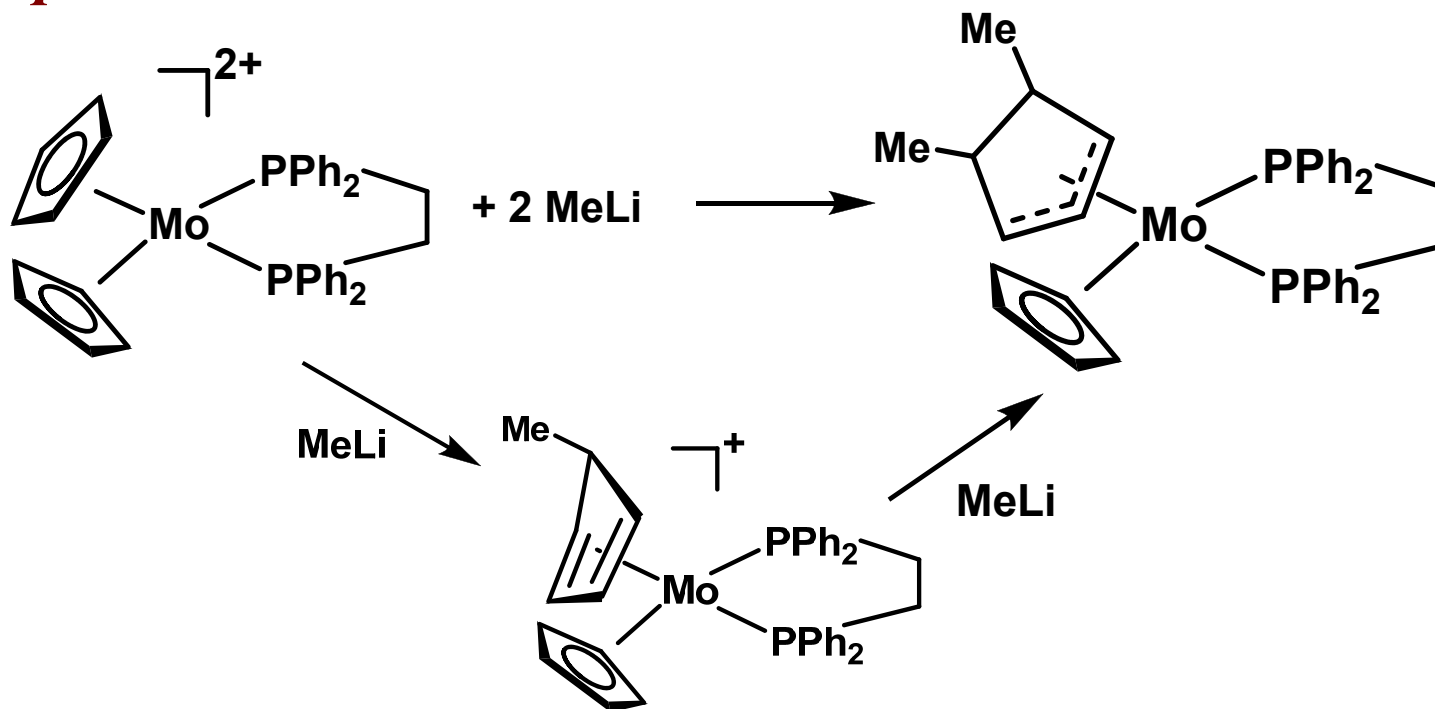


A rare example of attack at a Cp ring

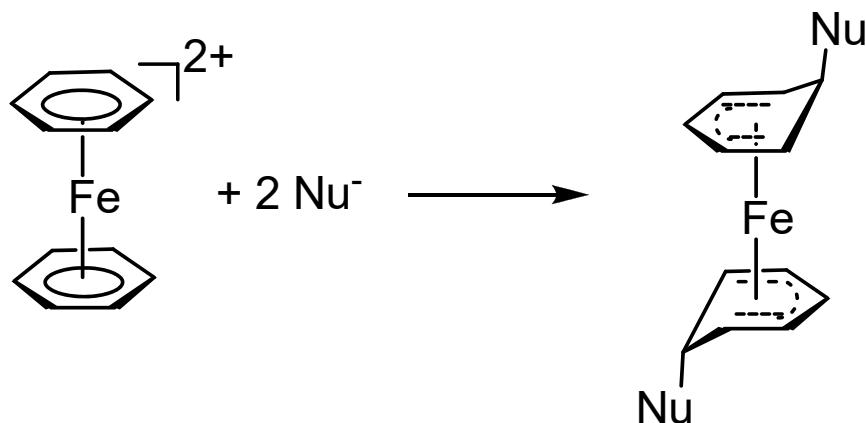
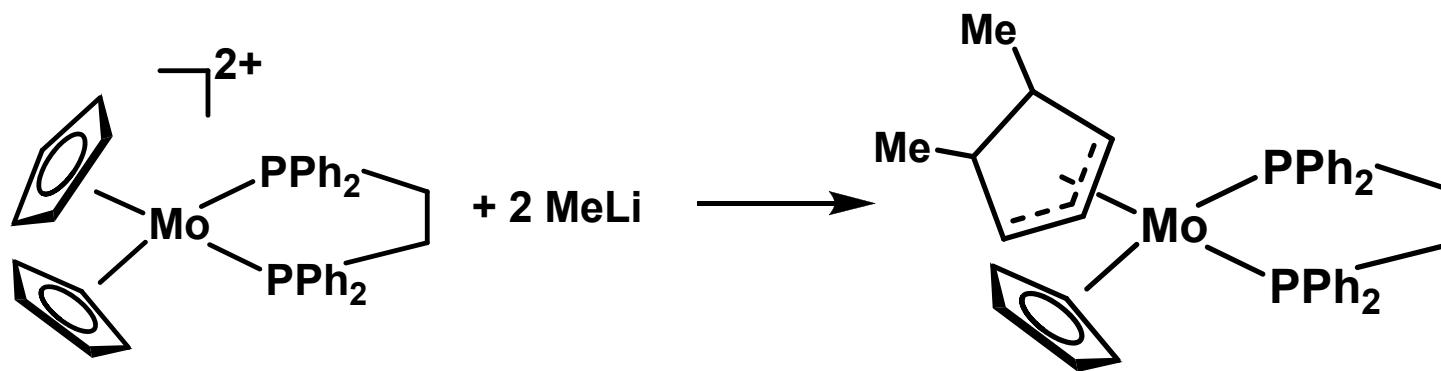
In general, Cp is a stabilizing ligand in studies on nucleophilic attack.

As an **odd closed system**, Cp is usually very resistant to attack and therefore directs addition to other ligands present on the metal.

Attack at a Cp ring only happens if there is no other π -bonded ligand present.



Rule 4. “Zeroth rule” of nucleophilic addition



A second nucleophile can also add, but should obey Rule 1.

“zeroth rule” of nucleophilic addition: *A nucleophile usually adds once to a monocation, twice to a dication, and so on.*

Qualitatively understanding these rules

虽然由经验发展而来，但分子轨道理论显示，这些规则经常能成功地预测LUMO轨道的主要位置。在动力学控制下,可以预期加成发生在空的接受轨道最集中的地方. 可以按如下所述定性地理理解这些规则

Qualitatively, we can understand the rules as follows:

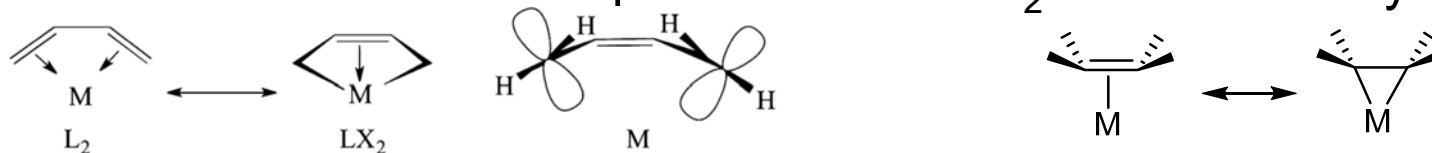
- Ligands having a higher **X** character tend to be more negatively charged and therefore tend to resist nucleophilic attack over **L** ligands. $\Rightarrow$ *Polyenes reacts before polyenyls*

e.g. A coordinated allyl, as an **LX** ligand, has more anionic character than ethylene, an **L** ligand.

- Can predict the relative reactivity of different ligands in the same class, a point not covered by the original rules. *e.g.*

Pentadienyl (**L₂X**) reacts before allyl (**LX**) because the former has the least X character.

Ethylene reacts before butadiene, as the **LX₂** form is always a more significant contributor to butadiene complexes than the X₂ form is for ethylene.

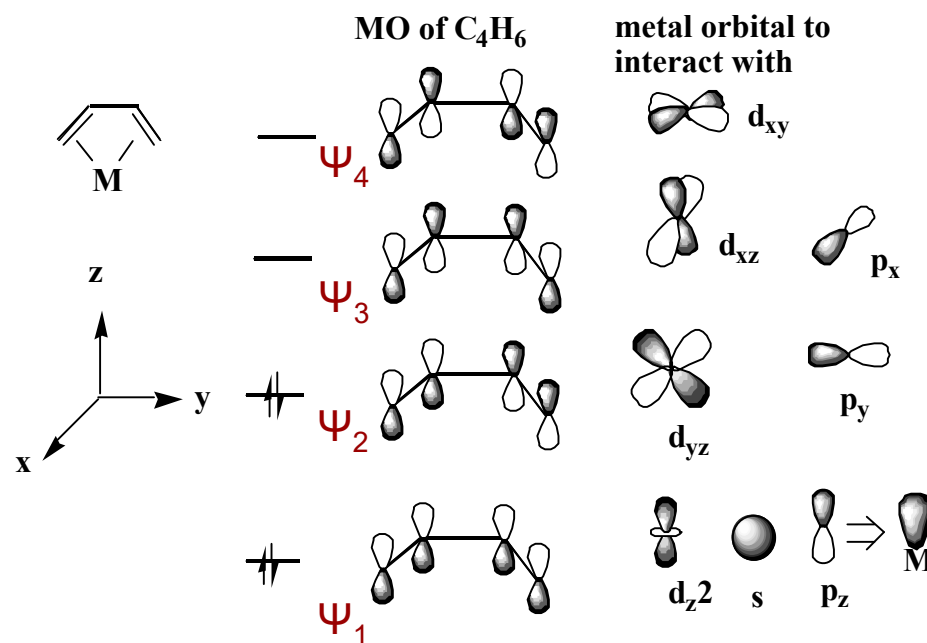


Qualitatively understanding these rules

- The terminal carbons of even, open ligands are the sites of addition.
 (偶电子、开放式配体的末端碳是发生加成反应的位点，因为LUMO轨道(Ψ_3)在该位置的系数较大)

MOs of butadiene and their interaction with metal

The LUMO (Ψ_3) is predominant at terminal carbon



Ψ_2 (HOMO) and Ψ_3 (LUMO) dominate bonding to the metal

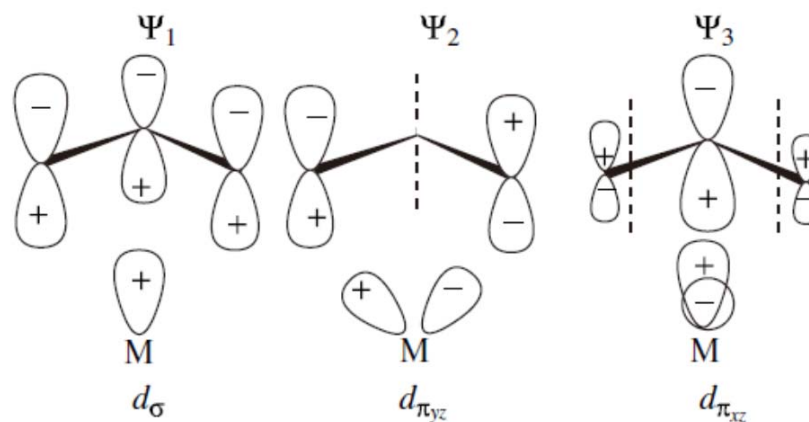
(For the electronic structure of butadiene, an electron-rich metal tends to populate Ψ_3 ; an electron-poor metal tends to depopulate Ψ_2).

Qualitatively understanding these rules

- An odd, open polyenyl gives terminal addition only if the metal is sufficiently electron withdrawing.

(奇电子开放式多烯基，在金属具有足够的吸电子能力时，才末端碳上发生加成反应)

Orbital interactions in ally complex



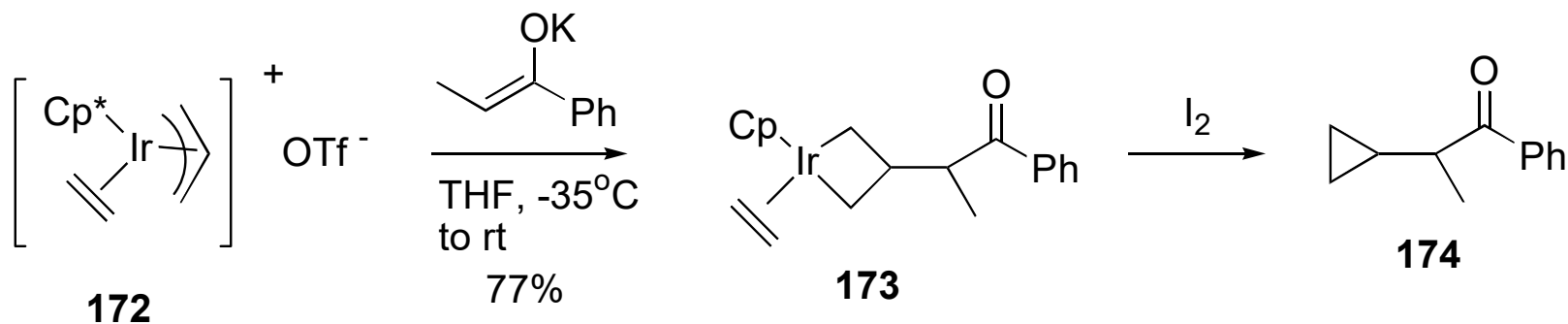
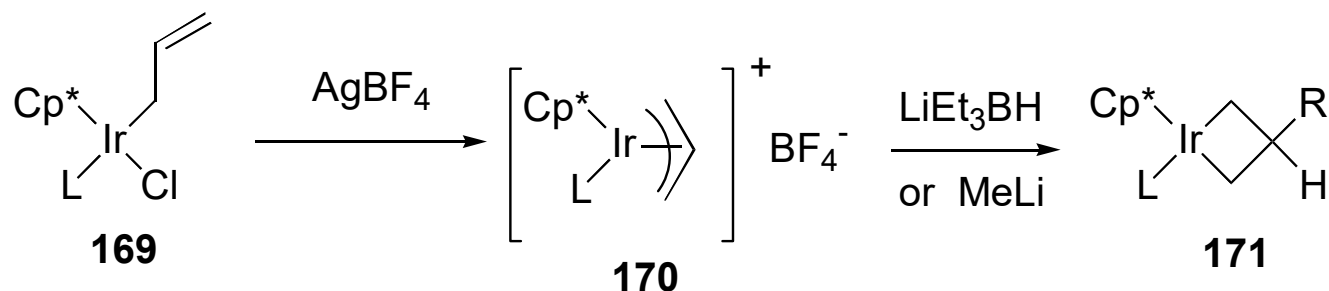
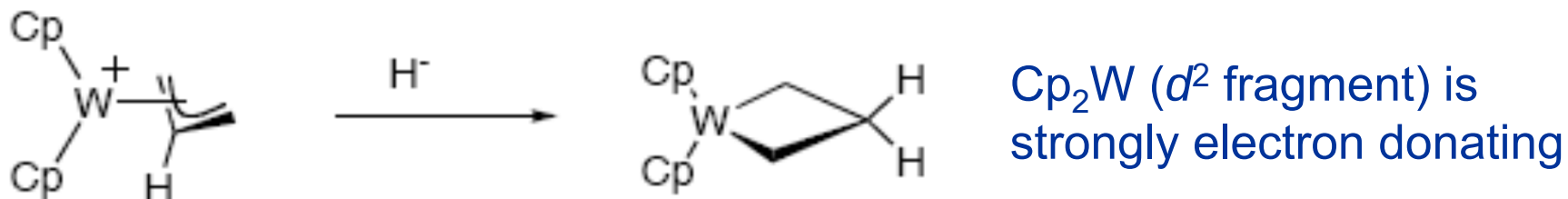
The LUMO of the free allyl ligand, Ψ_2 , prefers the terminus, but Ψ_3 is predominant at the central carbon. (自由配体的LUMO轨道 Ψ_2 倾向于在末端碳上，但是 Ψ_3 轨道主要集中在中间的碳上)

For a less electron-withdrawing metal, it tends to fill Ψ_2 , and to the extent that Ψ_3 becomes the new LUMO, we may no longer see terminal attack.

Some exceptions

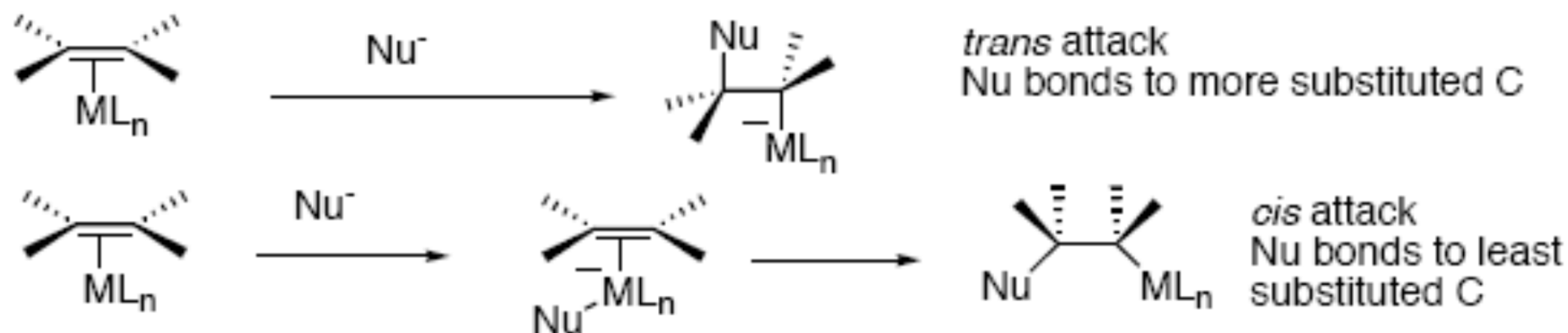
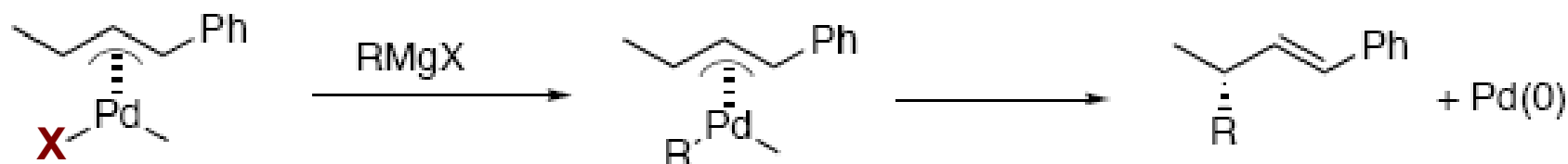
For allyl, the attack on central carbon may also occur.

Particularly when the L_nM is strong electron donating fragment.



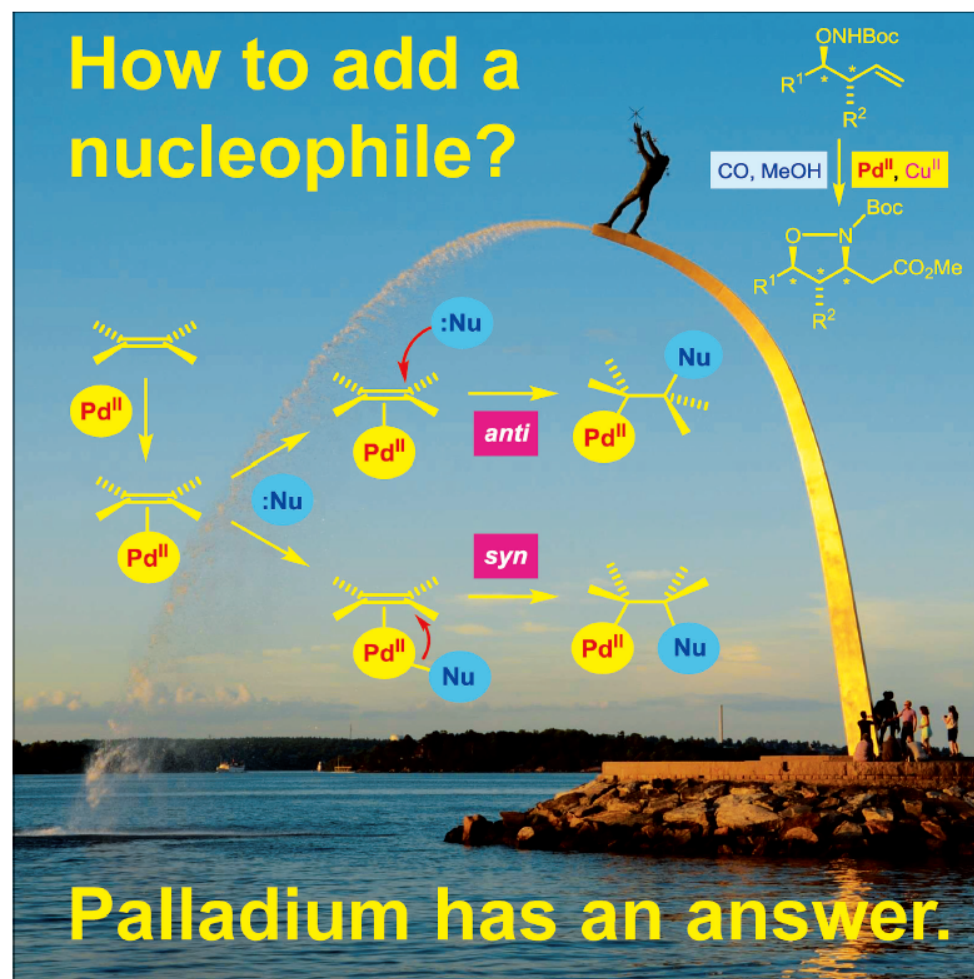
Some exceptions

Soft nucleophiles may also attack on the metal Initially followed by migratory insertion. e.g.



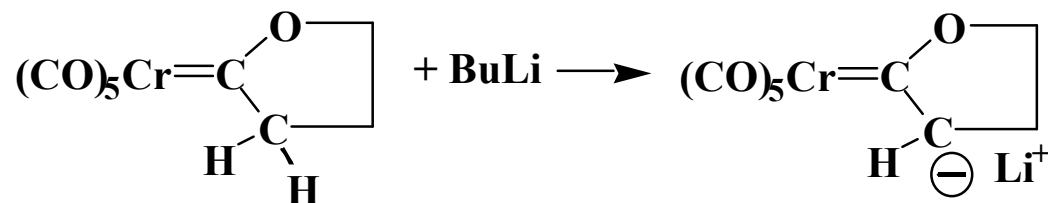
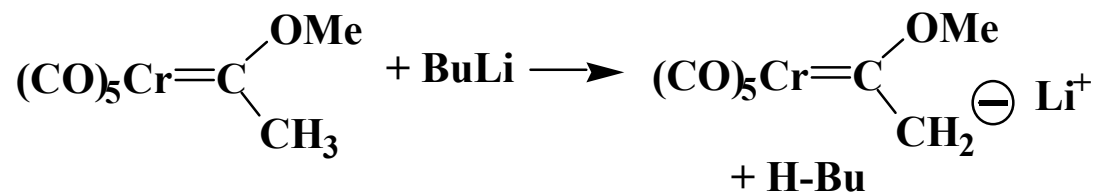
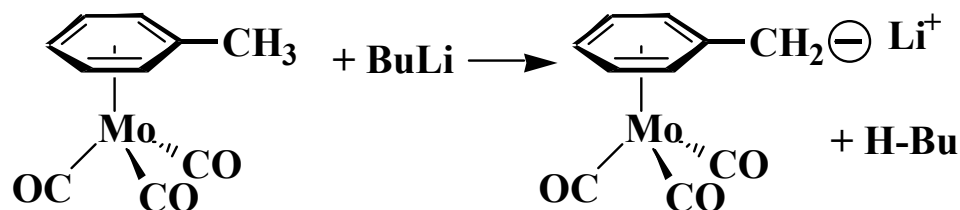
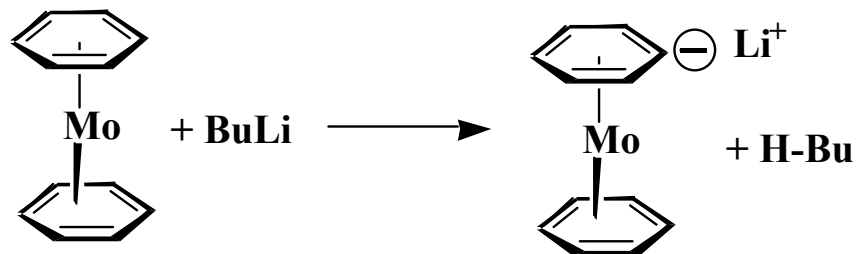
The syn/anti-Dichotomy in the Palladium-Catalyzed Addition of Nucleophiles to Alkenes

Pavel Koc̣ovský and Jan-E. Bäckvall*



D. Nucleophilic abstractions.

Organometallic complexes with acidic protons can be deprotonated. e.g.

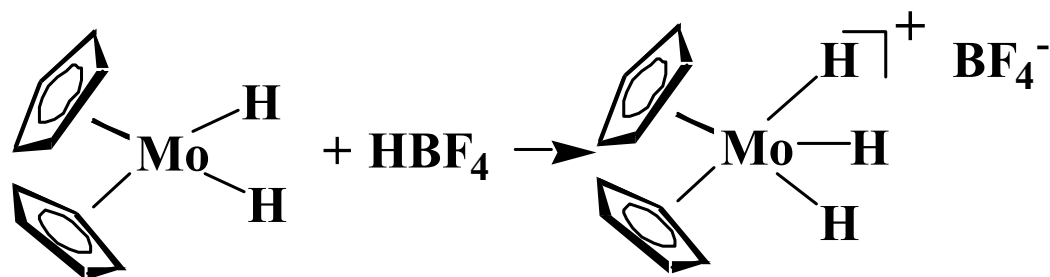
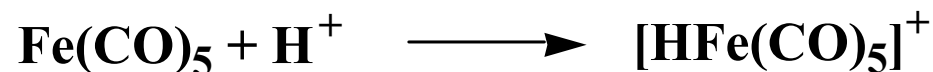
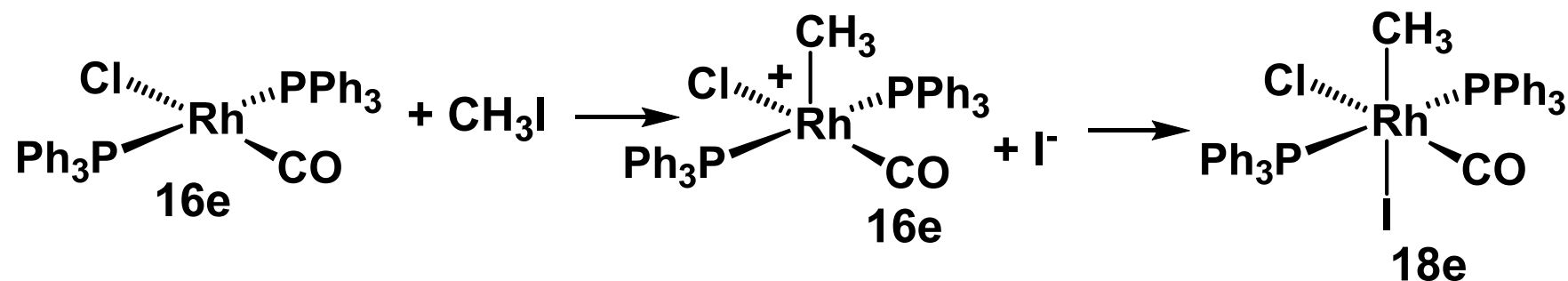


E. Electrophilic addition

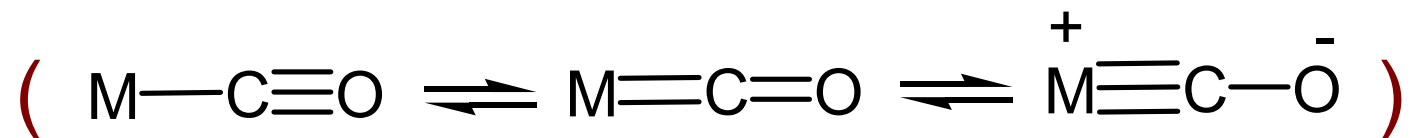
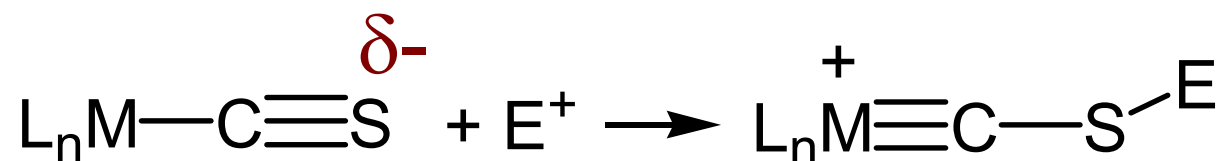
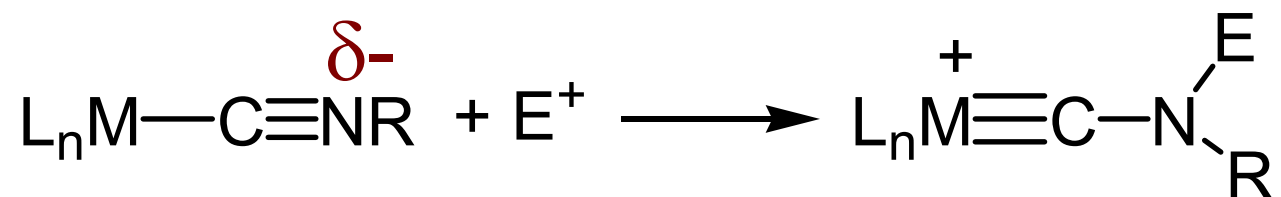
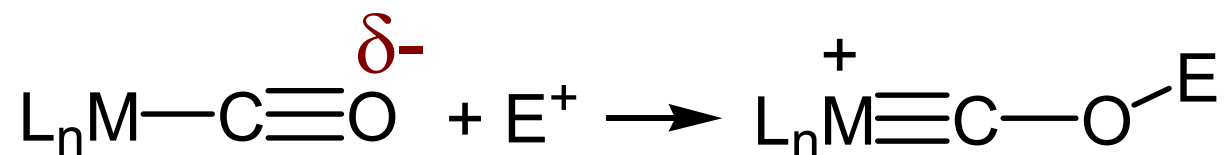
(1) Electrophilic addition to metals. (Special oxidative addition reaction)



e.g.

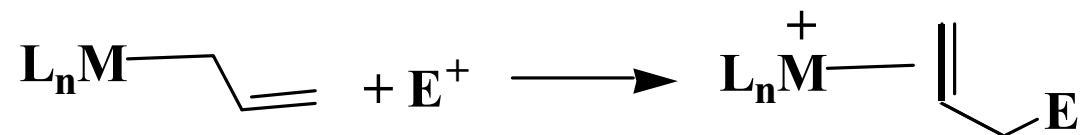
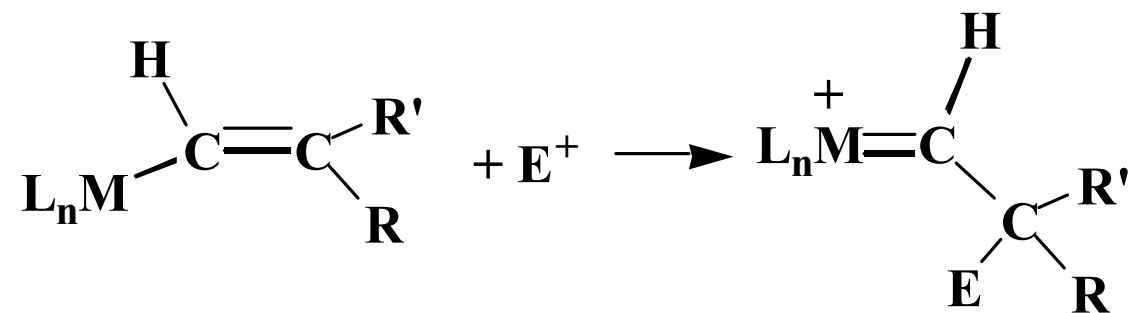
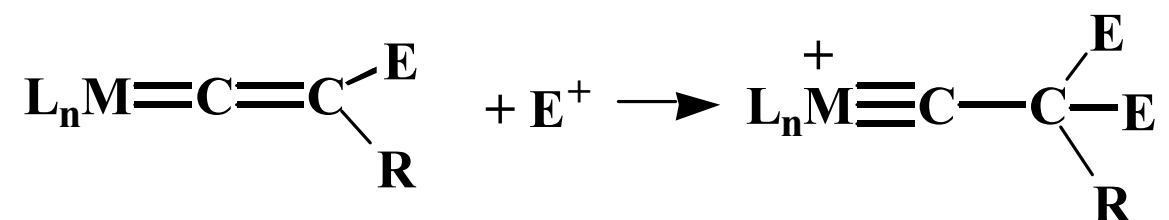
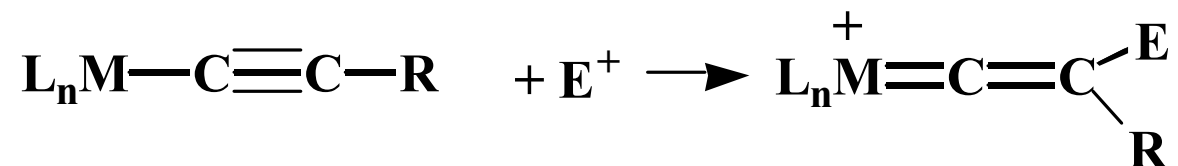


(2) Electrophilic addition to ligands.



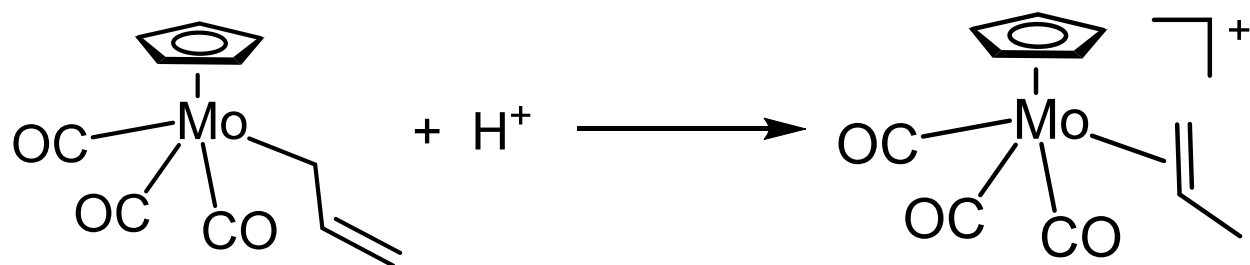
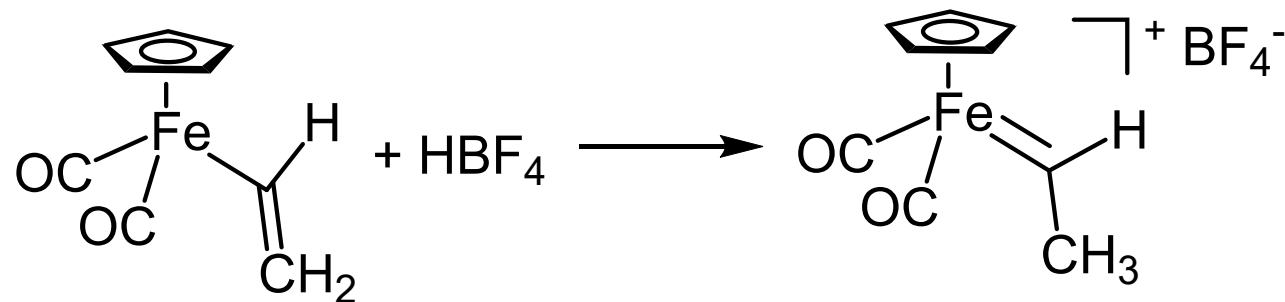
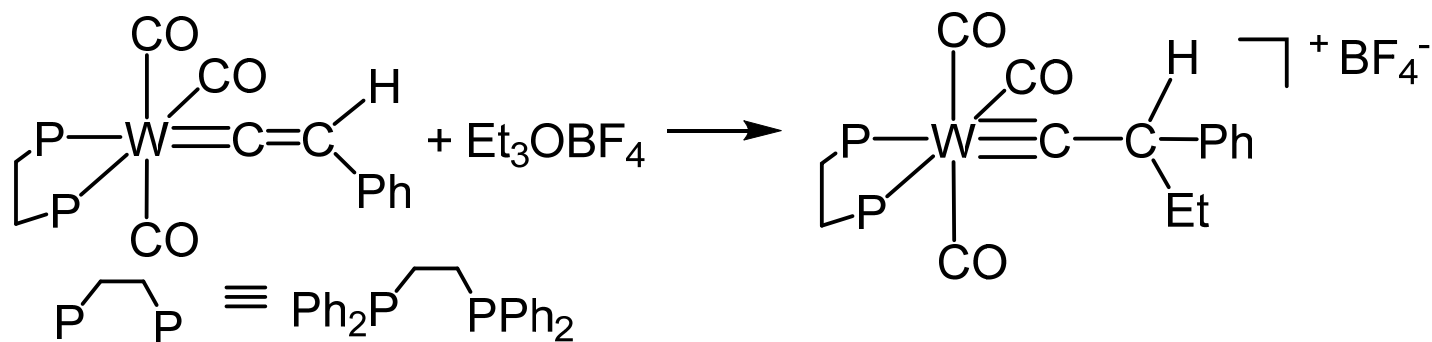
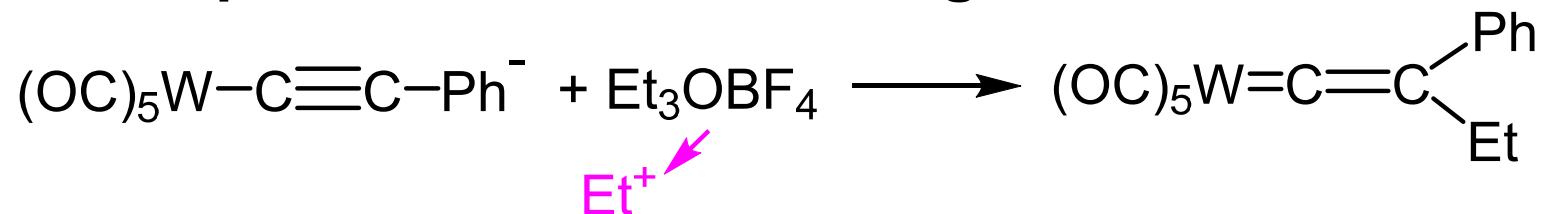
(Can be used for the synthesis of Fisher carbyne)

More examples



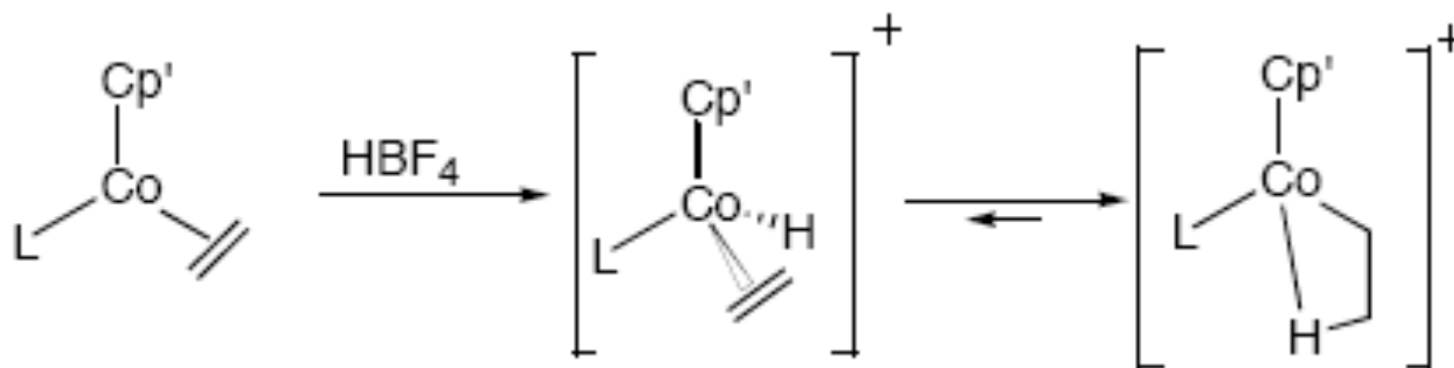
Practice Problems

Give the products of the following reactions.



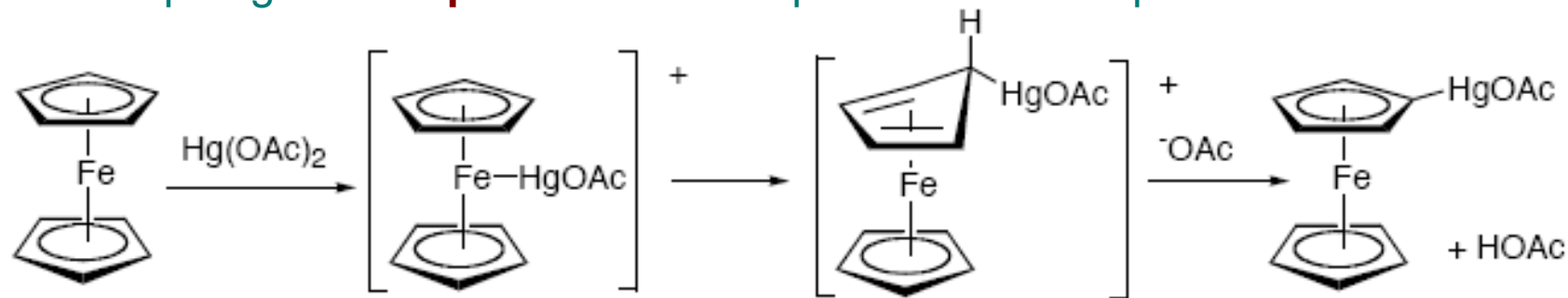
Comments on electrophilic addition reactions

Attack on olefin. The electrophile can either add first to the metal center, or directly to the ligand. The order of addition will depend on the relative basicity of the metal and the ligand.

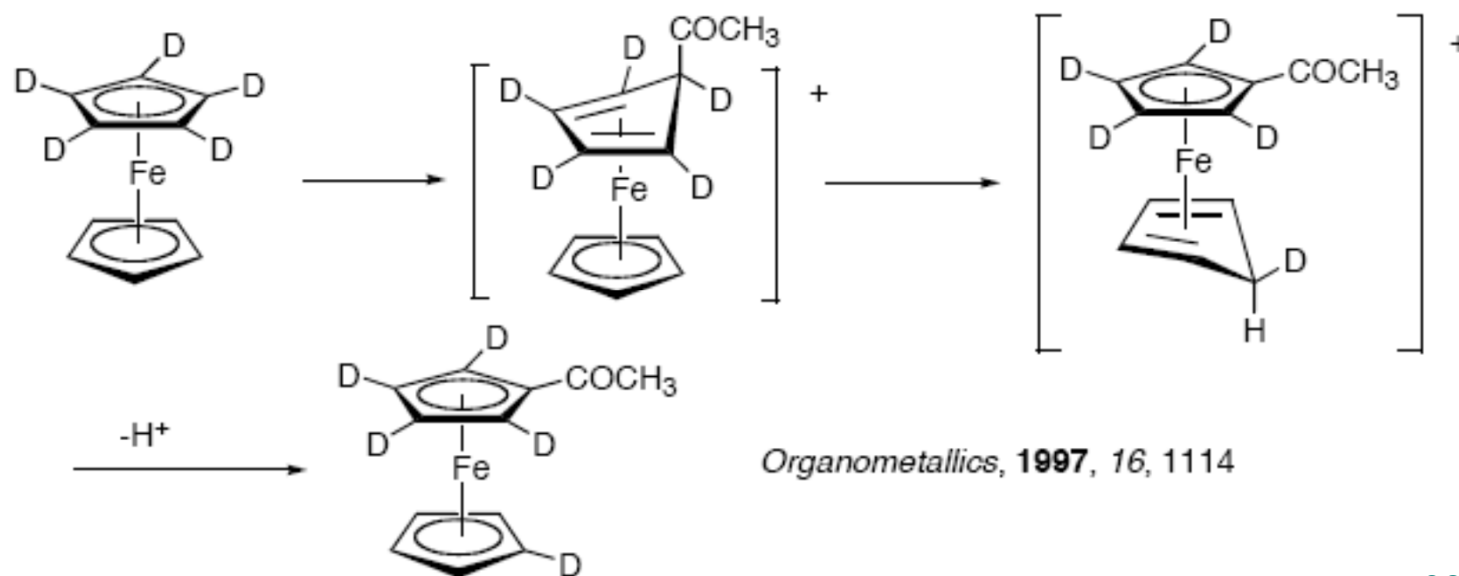


Electrophilic substitution of Cp ligand. Depending on the electrophile, addition to a Cp ring of ferrocene can either occur directly or by initial attack at iron.

Mercuric acetate adds **initially to the iron center** followed by **an endo** attack on the Cp ring. **Exo deprotonation** completes the electrophilic substitution.

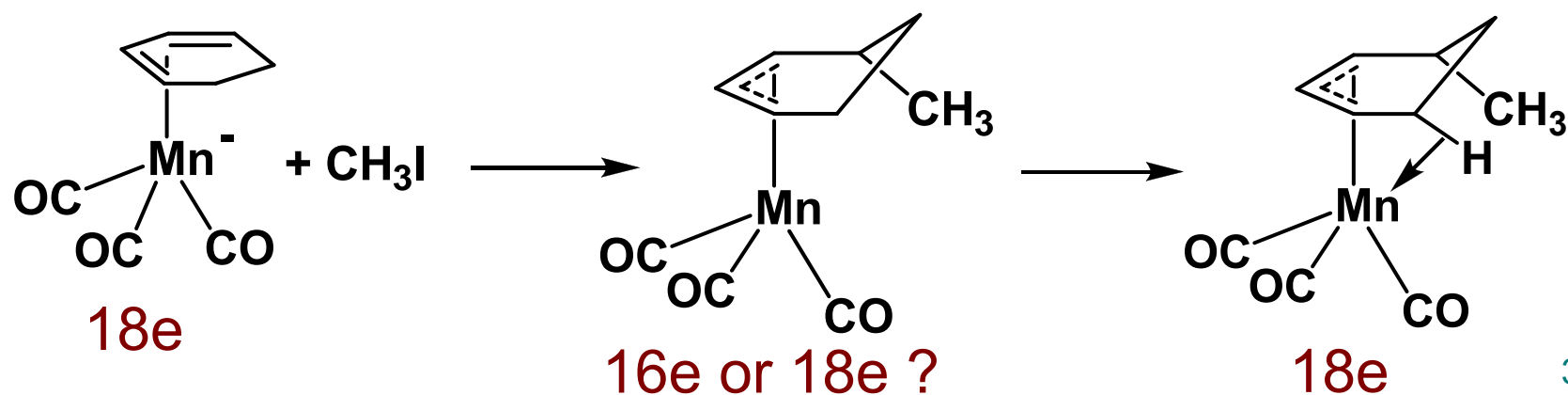
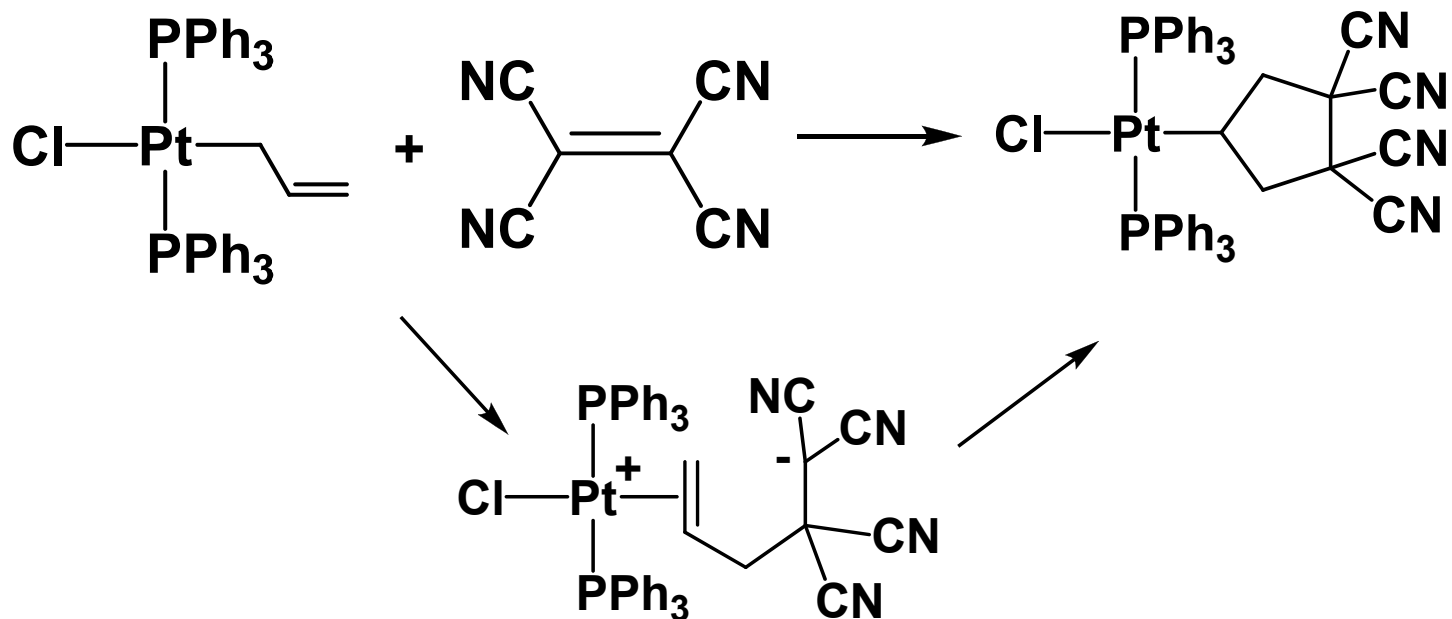


For a **hard electrophile**, such as CH_3CO^+ , **exo attack on the Cp ring** occurs directly



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The initially formed complex may undergo further transformation.
e.g.

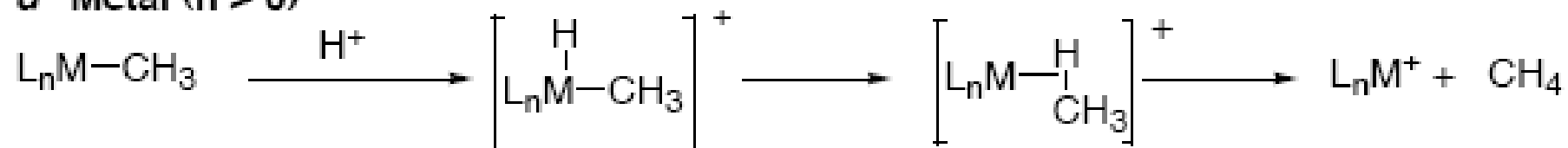


+

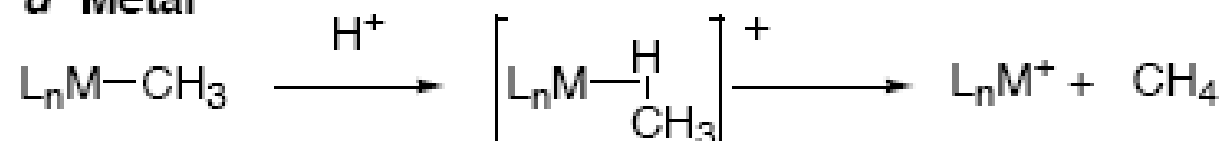
$$E_0 + \mathbb{H}C \longrightarrow$$

For metals with d electrons, protonation at the metal followed by reductive elimination is thought to occur. Since **d^0 metals cannot be protonated, attack must occur at the ligand.**

d^n Metal ($n > 0$)



d^0 Metal



Early transition metals are very reactive towards even weak acids such as water, but later transition metals require more vigorous conditions.

(2) Electrophilic abstraction of other groups.

Some groups can be electrophilically abstracted. e.g.

